

Measuring Habit Formation*

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Abstract

Habit formation is recognized as essential across numerous disciplines, yet economic measurement remains limited, fueling controversies over its existence, parameterization, and implications. This paper introduces direct micro-level measurements of habit formation through survey experiments. The findings reveal that habit forms internally, decays by two-thirds annually, and exerts a welfare impact equal to 63% of consumption's impact and triple that of peer effects. This decay estimate is precise relative to the range used in habit-based models, and contrary to common belief, habit formation alone (or peer effects alone) cannot resolve the Easterlin paradox, although combining them can largely account for it.

Keywords: Habit formation, consumption, micro evidence, Easterlin paradox, measurement

JEL codes: E21, G12, D15, I31, C81

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“The difficulty [in composing *The General Theory*] lies, not in the new ideas, but in escaping from the old ones[—‘habitual modes of thought and expression’], which ramify, for those brought up as most of us have been, into every corner of our minds.”

John Maynard Keynes, 1936

1 Introduction

For more than two thousand years, habit formation has been recognized as a fundamental determinant of human behavior across disciplines such as philosophy, psychology, biology, medicine, marketing, and education, and among prominent thinkers stretching from Aristotle and Confucius to Smith and Keynes. In economics, it has become a staple in macroeconomic and financial theory.¹ Yet after three decades of empirical work, economists still lack consensus on its most basic features: whether consumption habit exists (Cochrane, 2017), how fast it decays, and how much it matters for welfare. Thin micro evidence raises overfitting concerns that undermine model credibility (Breedon et al., 2015) and slows the further development of habit-based models for macro-financial analysis, where robust micro evidence serves as a foundation for predictions and policymaking (Stiglitz, 2018; Lucas, 1976).

The obstacle is measurement. Economic theories of habit formation center on how past total consumption affects current utility, but neither the habit nor utility is directly observed, and observed intertemporal choices also reflect income, wealth, expectations, and other constraints. Existing measurements are correspondingly limited: they focus on habit’s existence and intensity while largely neglecting its decay speed and welfare impacts, and they often infer from observational data under tenuous identifying assumptions, such as restrictive parametric structures or partial-spending proxies for total spending. Generating sustained variation in total spending to pin down these features is prohibitively costly in laboratory and field settings.

¹The recognition spans intellectual traditions. Aristotle observed that “moral virtue comes about as a result of habit” (*Nicomachean Ethics* 2.1), and Confucius that “what nature put together, habit separates” (*The Analects* 17.2). Smith observed in *The Wealth of Nations* (1776) that “the difference between the most dissimilar characters, between a philosopher and a common street porter, seems to arise not so much from nature, as from habit, custom, and education,” and Keynes argued in *The General Theory* (1936) that “a man’s habitual standard of life usually has the first claim on his income . . . when employment falls to a low level, aggregate consumption will decline by a smaller amount than that by which real income has declined, by reason both of the habitual behavior of individuals and also of the probable policy of governments.” Habit formation has since been incorporated into modern economic theory, with early contributions by Duesenberry (1949), Pollak (1970), and Spinnewyn (1981) in microeconomics; Ryder and Heal (1973), Carroll et al. (2000), and Fuhrer (2000) in macroeconomics; and Abel (1990), Constantinides (1990), and Campbell and Cochrane (1999) in finance.

This paper introduces an approach that can directly measure key aspects of the general habit-formation functional $u(C, H)$, overcoming the identification challenges in existing methods. In particular, we measure habit’s existence by assessing whether habit, H , defined as a function of past total consumption (C), reduces utility ($u_H < 0$)—the standard sign convention for habit formation in economics, as opposed to $u_H > 0$ (such as durability) or $u_H = 0$ (time-separability).² We also measure habit’s decay speed by gauging the speed at which H changes over time, and habit’s welfare impacts by measuring how H affects u . Under minimal restrictions on $u(C, H)$ —monotonicity in C ($u_C > 0$), differentiability, and exponential time discounting—we derive separate identification of these parameters through total consumption variations, the type of variation most directly informative about total-consumption habit formation.³ In principle, the identifying variations are implementable through any approach that can isolate them. We implement them through hypothetical choice survey experiments in order to circumvent the restrictive assumptions often required to isolate habit formation through observational data and feasibly achieve identification that remains elusive in laboratory and field experiments, while broadening the scope of evidence on habit formation in the literature.

Although hypothetical choices tend to raise validity concerns, this type of data is well-grounded in economics (see, e.g., Barsky et al., 1997; Manski, 2004; Ameriks et al., 2020; Stantcheva, 2023) and, when carefully designed, aligns closely with real-world decisions (Hainmueller et al., 2015; Matousek et al., 2022; Kumar et al., 2023; Gneezy et al., 2024), particularly in spending contexts (Coibion et al., 2024). Unlike traditional real-world data, which often suffer from identification issues, our implementation follows the “data engineering” strategy—generating new forms of data closely linked to model constructs (Caplin, 2025, 2024; Almås et al., 2024)—which has been applied to expectations (Coibion et al., 2022; Andre et al.,

²Our identification operates at this functional level: “habit formation exists” is a claim about the sign of u_H , and mechanisms that generate the same functional—such as backward-looking reference dependence—are observationally equivalent and interchangeable for our purposes (see Section 2.1). Other alternatives, including expectations-based reference dependence and preferences over consumption sequences, carry different implications for response patterns the survey reports; Section 3.1 examines them.

³We focus on total-consumption habit formation because it is the type of habit formation most directly tied to the macroeconomic and financial applications that have made habit formation important in economics. This differs from deep-habit models, where habits attach to particular goods or varieties (Ravn et al., 2006), and from models of cue-based or automation-based choice (Matyskova et al., 2020; Camerer et al., 2024); those models are important but do not directly discipline the total-consumption functional studied here. The cross-partial derivative $u_{CH} > 0$, often associated with habit formation in the literature (e.g., Constantinides, 1990), is implied by $u_H < 0$ under canonical habit specifications with standard parameter values. We defer measurement of such second-order derivatives to a companion paper.

2022), earnings (Weidmann and Deming, 2021), consumer spending (Ameriks et al., 2007; Coibion et al., 2024), and firm decisions (Kumar et al., 2023). Recent macro-finance work also illustrates how micro-level data can discipline latent behavioral mechanisms in aggregate models (e.g., Bordalo et al., 2026). In the present setting, hypothetical choices offer two advantages: they can identify habit parameters that remain elusive under traditional methods, and they provide complementary evidence for parameters traditional methods can also address, since hypothetical bias tends to be orthogonal to the identification challenges that limit traditional methods. To strengthen validity, we anchor spending variations in empirical realities, use a two-wave design to mitigate response error, and add experimental controls that isolate habit formation from confounds such as durable goods. Our findings are corroborated by respondents’ own explanations and by related behaviors in an independent dataset and the literature, exhibiting predictive and convergent validity.

Using this approach, we document new micro evidence on habit formation, covering its existence, decay speed, and welfare impacts. First, habit forms internally, based on an individual’s own past consumption; we find no evidence that it also forms externally, based on others’ past consumption. This finding provides direct evidence supporting the common preference-based approach to modeling habit formation, especially for those internal-habit formulations. Second, habit decays at a rate of two-thirds per year, a precise estimate relative to the wide range used in habit-based models. Third, habit formation has a substantial welfare impact, equivalent to 63% of consumption’s impact and three times that of peer effects. Finally, contrary to common belief, habit formation alone—or peer effects alone—cannot explain the Easterlin paradox, but their combination provides a plausible explanation.

To obtain these results, we use a general model that is agnostic about whether habit formation exists and how it enters utility, while nesting standard models with habit formation. This generality matters because rejecting a specialized version does not refute the general idea of habit formation (Cochrane, 2017). We then specify preference parameters governing the existence, decay speed, and welfare implications of habit formation (Section 2.1), and identify them through total-consumption variations implemented in a survey on Prolific, yielding a broadly representative U.S. sample of about 2,100 responses.

Following detailed instructions and comprehension checks, the survey presented respondents with a series of binary choice questions asking about the (utility) ranking of spending profiles. The profiles

differ in the timing and level of spending, with both dimensions anchored to ensure familiarity. Such hypothetical variations are routinely used to identify preference parameters in the literature (e.g., Barsky et al., 1997; Ameriks et al., 2020). Leveraging humans’ natural capacity for counterfactual thinking about both past and future variations (Epstude and Roese, 2008), the differing spending profiles induce variations in habitual spending levels, thereby influencing the utilities of the profiles. The ranking of the utilities is thus a function of the spending difference and the preference parameters. Given the spending difference, inverting the utility ranking indicated in the survey responses recovers the preference parameters, a procedure we formalize in Section 3. Section 2.2 details the design of our survey experiments, such as the visual presentation of the spending profiles and the hypothetical scenario for reducing cognitive load and controlling for confounding factors. Section 2.3 discusses our strategies to enhance the validity of our hypothetical choices and presents evidence of their effectiveness. Section 2.4 summarizes the sample. Section 2.5 presents the statistical model for extracting reliable responses from the two survey waves while controlling for response errors not addressed by the survey experimental design.

Building on this framework, Section 3 details the measurement of the preference parameters of interest, starting with the existence of habit formation (Section 3.1). Because internal and external habit formation can have different implications for optimal tax policy and welfare analysis (Carroll et al., 1997; Ljungqvist and Uhlig, 2000), we measure them separately. To identify internal habit formation, we vary own past spending while holding future spending fixed. Choosing the profile with the lower past spending for a more satisfying future experience implies $u_H < 0$ and therefore internal habit formation. About two-thirds of respondents made this choice in each wave, and estimation accounting for response errors and respondents’ open-ended explanations both confirm internal habit formation. A parallel question on external habit formation, varying only other people’s past spending, provides no evidence supporting its existence.

Given the existence of habit formation, we next measure the speed of habit decay (Section 3.2), which is central to habit-based models but has received far less empirical attention than habit intensity. Because the decay rate determines how quickly the effect of past consumption dissipates, we identify it by varying the persistence of past spending: if sustained past spending affects utility less than recent transitory past spending, habit decays faster than a threshold (Proposition 1). The estimated decay speed is about

two-thirds per year, with a confidence interval ([0.58, 0.69] at the 95% level) that is narrow relative to the range used in habit-based models—from full decay after a quarter in many macroeconomic applications (e.g., Smets and Wouters, 2007; Christiano et al., 2005) to about 11% per year in Campbell and Cochrane (1999). Section 4.1 illustrates the quantitative importance of the decay speed in a widely cited habit formation framework. As validation, we show that faster habit adjustment, as measured by our approach, predicts faster real-world hedonic adaptation in an independent dataset collected two months after ours.

Section 3.3 details the measurement of two preference parameters that govern the welfare impacts of habit formation. In particular, the slope of the indifference curve, $-\frac{u_H}{u_C}$, measures the welfare impact of habit formation relative to that of one's own consumption. The ratio $\frac{u_{C_{\text{others}}}}{u_H}$ assesses the relative welfare impacts of habit formation and peer effects from others' contemporaneous consumption (C_{others}). Both parameters offer insights into the resolution of the Easterlin paradox. As ratios, the identification of these parameters naturally depends on trade-offs: habit versus consumption (for $-\frac{u_H}{u_C}$), and peer effects versus habit (for $\frac{u_{C_{\text{others}}}}{u_H}$). Leveraging these tailored identifying variations, our findings imply a habit intensity of approximately 0.6, providing direct micro-level evidence for a value often used in macroeconomic calibrations. Furthermore, habit formation's welfare effect is found to be three times that of peer effects.

Section 4 presents two applications of our measurements of habit formation. Section 4.1 examines what the measured habit decay implies for the equity premium puzzle (Mehra and Prescott, 1985), a cornerstone application of habit formation. Applied to the canonical Campbell and Cochrane (1999) habit-formation model, our estimated annual habit decay factor of 0.64 generates an expected excess return of 41% and a Sharpe ratio of 1.35—more than six times the postwar excess return of 6.7% and three times the postwar Sharpe ratio of 0.43. Matching the observed moments requires a habit decay factor well outside the 99% confidence interval of our estimate. Because Campbell and Cochrane (1999) calibrate habit persistence to the serial correlation of the log price-dividend ratio, an aggregate time-series moment rather than a preference target, the discrepancy indicates that the slow habit decay in that calibration reflects asset-pricing fit more than the underlying preference parameter.

Section 4.2 investigates the potential of habit formation to explain the half-century-old Easterlin paradox: happiness and income are positively correlated in the cross-section and short run but essentially uncor-

related in the long run (Easterlin, 1973, 1974; Stevenson and Wolfers, 2008; Easterlin, 2017; Easterlin and O'Connor, 2025). Although habit formation has long been proposed as an explanation (Easterlin, 1995; Clark et al., 2008), direct evidence on whether it can actually account for the paradox has been absent. Using this paper's measurements, we find that neither habit formation nor peer effects—another popular explanation—alone can fully account for the happiness–income dynamics. Combining the two, however, can account for much of the observed pattern across all three dimensions: they erode long-run happiness gains from income, preserve short-run co-movement while adjustment is incomplete, and generate a positive cross-sectional gradient through the accumulation of transitional gains. The mechanism implies that, despite the negligible long-run happiness–income gradient, sustained economic growth remains important for maintaining the stabilized level of happiness.

Section 5 concludes.

Related literature. The primary contribution of this paper is a methodology for directly measuring habit formation. It adds to four literatures: micro evidence on habit formation, the Easterlin paradox, reference dependence, and structural preference elicitation.

Over the past three decades, the study of micro evidence on habit formation has become an active field. Using mostly observational data, it centers on the existence of internal habit formation and yields divergent findings: Dynan (2000), Meghir and Weber (1996), and Brunnermeier and Nagel (2008) find no evidence, while Crawford (2010), Naik and Moore (1996), and Khanal et al. (2018) find support, and Browning and Collado (2007) report mixed results. The limited evidence on external habit formation is likewise conflicting, as Dynan and Ravina (2007) support its presence while Brunnermeier and Nagel (2008) do not. Moreover, habit intensity estimated from micro data often falls short of the levels required to account for macroeconomic data (see Havranek et al., 2017, for a review).

Experimental studies more consistently find evidence of habit formation, though often in non-consumption settings—gym attendance and handwashing (Charness and Gneezy, 2009; Royer et al., 2015; Hussam et al., 2022), social-media posting and tuna purchases (Camerer et al., 2024), and state-recognition tasks (Matyskova et al., 2020). Hypothetical intertemporal choices in Kapteyn and Teppa (2003) favor a multiplicative habit model over a non-habit alternative.

This paper contributes to the literature by developing identifying variations that can directly measure total-consumption habit formation. Using the recently termed “data engineering approach” (Caplin, 2025, 2024; Almås et al., 2024), our survey implementation of the variations isolates habit formation from confounding factors in observational data and avoids the prohibitive costs of studying total-consumption habit formation in laboratory and field settings, addressing concerns due to heterogeneity of habit formation across behaviors (Wood and Rünger, 2016; Buyalskaya et al., 2023). The validity of these identifying variations rests on standard economic assumptions, independent of the survey implementation. This approach yields direct micro-level evidence supporting the widely used preference-based framework for modeling habit formation, while addressing concerns in the existing literature, such as the reliance on partial spending data, parametric assumptions, and model fit over direct examination of habit’s effect on preference structure. Using a hypothetical scenario with minimal adjustment costs, our approach provides evidence on habit formation that complements Chetty and Szeidl’s (2016) consumption-commitment account. By tailoring identifying variation for each of the habit-related parameters, this paper broadens the scope of micro evidence on habit formation, enabling investigation of habit’s decay speed and welfare impacts relative to consumption and peer effects, with fewer constraints from specification and data availability.

A more focused literature tests habit formation models while relaxing parametric assumptions: Chen and Ludvigson (2009) let habit evolve nonparametrically under additive habit and power utility specifications. Crawford (2010) relaxes parametric assumptions on both utility and habit evolution. We go further with a nonparametric utility function that relaxes joint concavity in consumption and habit, thereby encompassing the common multiplicative habit specification. We further accommodate infinitely many lags in habit evolution and leverage data specifically designed to measure total-consumption habit formation.⁴

In the five decades since the Easterlin paradox was introduced (Easterlin, 1973, 1974), many explanations have been proposed to resolve it, but how to validate them—especially those based on unobservable constructs such as habit formation (Easterlin, 1995; Clark et al., 2008)—remains an open question. This paper offers a perspective for tackling the issue: measuring the constructs with minimal restrictions and conducting calibrated quantitative investigations to assess the extent to which each explanation can

⁴In order to measure habit’s decay rate and welfare impacts, we specialize to a standard linear habit evolution (Section 2.1). For the existence of habit formation, we impose no restriction on the habit evolution beyond that past spending raises habit.

contribute to resolving the paradox. Using this approach, we corroborate habit formation’s key role in explaining the paradox, yet challenge the view that it alone suffices. This study further advances the literature by controlling for and assessing changes in reporting scales à la Benjamin et al. (2023), thereby differentiating the explanation through habit formation (and peer effects) from the rescaling explanation (Deaton, 2008; Prati and Senik, 2024). The paradox’s near-zero long-run happiness–income gradient has been used to critique growth-centric policies (e.g., Alexander, 2012); our explanation implies instead that sustained economic growth, though perhaps insufficient to elevate long-run happiness, is necessary to maintain it.

This paper also relates to the extensive literature on reference dependence (Markowitz, 1952; Kahneman and Tversky, 1979), where reference points are typically anchored to the status quo (e.g., Masatlioglu and Ok, 2014) or expectations (e.g., Kőszegi and Rabin, 2006). This literature has linked habit formation with reference dependence, leading to the introduction of backward-looking reference points (e.g., Bowman et al., 1999; Post et al., 2008). Consistent with existing evidence (e.g., Della Vigna et al., 2017), we provide direct evidence for backward-looking reference points in the total-consumption setting and quantify their evolution and welfare impact.

Finally, the paper contributes to structural preference elicitation. This literature has mostly focused on measuring risk aversion, time discounting, intertemporal substitution, and well-being trade-offs under fully parametric preferences (e.g., Barsky et al., 1997; Kimball et al., 2008, 2009, 2024b) or first-order approximations for semiparametric and nonparametric preferences (e.g., Benjamin et al., 2014b, 2019). We extend it to habit formation: by designing variations in past consumption, we identify habit decay speed exactly and its welfare impacts through first- and second-order approximations under a nonparametric utility function.

2 Methodology

2.1 Model and Preference Parameters

To address the gaps in the existing evidence, we use a general model that is agnostic regarding the existence and specification of habit formation, while retaining standard assumptions needed to speak to existing

models. Specifically, the agent maximizes

$$\int_0^{\infty} e^{-\rho t} u(C_t, H_t) dt,$$

where C_t is the agent's total spending, H_t is their habitual level of spending (henceforth, habit), and ρ is the time discount rate. Unless otherwise specified, the time index is omitted hereafter for brevity. We follow the literature in maintaining exponential time discounting and the restriction that past consumption influences current utility, if at all, only through the habit H . The nonparametric utility function u is assumed three times continuously differentiable and strictly increasing in consumption ($u_C > 0$). Beyond these assumptions, the functional form of $u(C, H)$ is left unrestricted; in particular, the analysis imposes no parametric form on whether and how habit enters utility. The utility function may depend on additional variables, such as leisure. Since these variables remain constant in the identifying variations, omitting them from the explicit utility function entails no loss of generality. When examining changes in variables beyond own consumption and habit—such as others' current consumption—these additional variables will be explicitly incorporated into the utility function.

For the existence of habit formation, we do not need to specify a habit evolution equation, except to require that habit rises with past consumption, a monotonicity shared by effectively all existing habit evolution equations. Quantifying habit's decay rate and welfare impacts, however, requires putting habit on a scale. We therefore assume habit evolves according to

$$\dot{H} = \theta (C - H), \tag{1}$$

where θ is the habit decay rate. This equation makes habit an exponentially weighted average of past spending, with the weight on each period's spending declining with the elapsed time at rate θ .

We use this specification for four reasons. First, it is the most widely used habit evolution equation in the literature, with the discrete-time expression $H_t = (1 - e^{-\theta}) C_t + e^{-\theta} H_{t-1}$. Researchers have used different habit evolution equations; however, their differences either amount to a simple rescaling of the unit of habit (e.g., the habit evolution equation of [Constantinides, 1990](#)) or vanish in the steady state (e.g., the habit evolution equation of [Campbell and Cochrane, 1999](#)). Second, the habit evolution equation can

accommodate mixtures of internal and external habits by replacing C with the relevant weighted average of own and others' consumption (e.g., [Grishchenko, 2010](#)), with negligible impact on the results presented in this paper. Third, this habit evolution equation provides an intuitive unit for habit, identical to that of consumption, consistent with H 's definition as the habitual level of spending. For example, a person who has spent \$5,000 per month for as long as they can remember has a habitual spending level of \$5,000 per month. More generally, with habit measured in this unit, smooth nonlinear habit evolution equations have equation (1) as their first-order expansion around steady state.⁵ Fourth, the exogeneity of this habit evolution equation is supported by extensive evidence from the biology literature showing habituation across diverse species, including single-celled organisms without neural systems ([Boisseau et al., 2016](#)).

These assumptions complete the model's specification. The model is intentionally parsimonious, allowing it to nest more specialized models whose extra structures are held constant in the experiments. Furthermore, we do not explicitly model budgetary constraints, as the respondents choose among complete, budget-feasible spending profiles (see Section 2.2 for details).

Our identification operates at the level of the functional $u(C, H)$ and is agnostic about the underlying mechanism. By "habit formation" we mean this functional, so "habit formation exists" is a claim about the sign of u_H , not a claim that habituation has been empirically distinguished from every alternative mechanism that can generate the same functional representation. This functional-level focus follows standard economic practice: preference functionals, rather than the psychological mechanisms that generate them, are the primitive objects of economic theory, and measuring them is the standard target of structural estimation and welfare analysis. The same functional can arise from a range of backward-looking mechanisms, including backward-looking reference dependence ([Bowman et al., 1999](#)), memory-based utility ([Mullainathan, 2002](#)), and consumption-dependent discounting ([Shi and Epstein, 1993](#)). For the choices and applications studied here, such mechanisms can be observationally equivalent and interchangeable; following [Crawford \(2010\)](#), we view distinguishing among them as beyond the paper's scope. We retain the label "habit formation" because the macroeconomic and financial literatures in which this functional plays a central

⁵To see this, write a smooth evolution as $\dot{\mathcal{H}} = f(C, \mathcal{H})$ with $f = 0$ in every steady state. Differentiating along the diagonal gives $f_C = -f_{\mathcal{H}}$ at any steady state, so the first-order expansion is $\dot{\mathcal{H}} = \theta(C - \mathcal{H})$ with $\theta = -f_{\mathcal{H}}$. For linear f this expansion is exact, so θ is the exact decay rate for the linear specifications used in the literature; for nonlinear evolutions, such as that of [Campbell and Cochrane \(1999\)](#), θ is a steady-state decay rate.

role recognize it under that name. Alternatives outside this equivalence class include, notably, expectations-based reference dependence (Kőszegi and Rabin, 2006) and preferences over properties of consumption sequences (Loewenstein and Sicherman, 1991; Loewenstein and Prelec, 1993), as well as response heuristics unrelated to preferences. Section 3.1 examines these against the observed response patterns.

To provide micro evidence on habit formation, it is essential to determine whether and how habit influences utility. Accordingly, we measure six relevant preference parameters: the existence of internal habit formation, the existence of external habit formation, the habit decay rate (θ), the time discount rate (ρ), and two utility-derivative ratios ($-\frac{u_H}{u_C}$ and $\frac{u_{C_{\text{others}}}}{u_H}$).⁶ The habit decay rate (θ) determines how rapidly habit adjusts to changes in consumption, while $-\frac{u_H}{u_C}$ captures the welfare impact of habit formation relative to consumption. $\frac{u_{C_{\text{others}}}}{u_H}$ quantifies the relative welfare impact of peer effects versus habit formation.

2.2 Survey Experimental Design

Our survey is structured as follows: (i) consent form; (ii) demographic questions; (iii) calibration questions;⁷ (iv) instructions on spending graphs with comprehension checks; (v) instructions on the hypothetical situation with comprehension checks; (vi) four modules on internal habit formation, habit decay rate, slope of indifference curve, and time discount rate; (vii) instructions on spending graphs under uncertainty; (viii) three modules on higher-order habit parameters;⁸ (ix) instructions on spending graphs with other people’s spending; (x) three modules on external habit formation, external habit mixture coefficient, and $\frac{u_{C_{\text{others}}}}{u_H}$; (xi) comprehension checks on the hypothetical situation (repeated); (xii) calibration questions (repeated); and (xiii) exit questions on respondents’ approach to the survey and willingness to participate in a follow-up wave. Appendix F presents screenshots of the full survey.

Past spending shapes habit, which in turn may influence well-being, suggesting that varied spending profiles can produce distinct welfare outcomes, with preference parameters modulating the process. Our

⁶Estimating certain preference parameters requires knowledge of the time discount rate, ρ , whose elicitation is relegated to Appendix E.1 given its indirect relevance.

⁷Calibration questions are survey questions designed to have the same objective state across all respondents, enabling the identification and measurement of scale-use heterogeneity across individuals (Benjamin et al., 2023).

⁸Modules (vii) and (viii), which measure second-order utility derivatives, are designed to address distinct research questions (e.g., testing additive and multiplicative habits) and are deferred to a companion paper. Module (x) includes questions to measure an external habit mixture coefficient, which is omitted from this paper given evidence on the absence of external habit formation (Section 3.1).

Remember, future experience reflects how you feel about the future starting now.
Which universe will provide you with a more satisfying **FUTURE** experience?

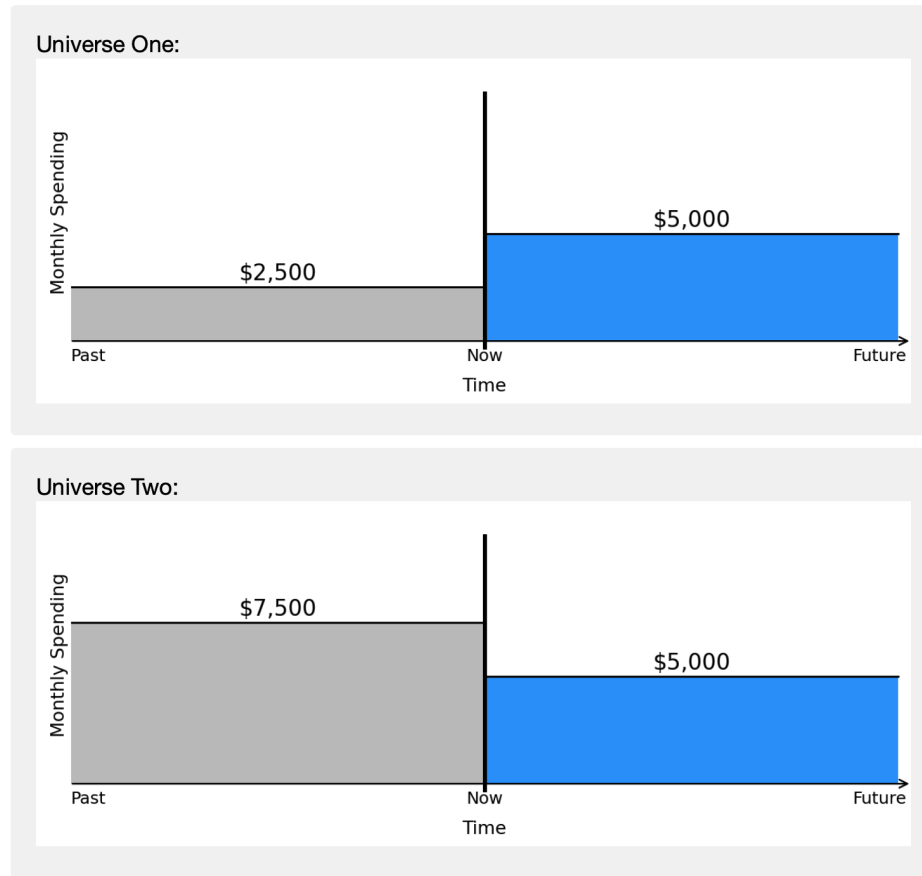


Figure 1: A Typical Survey Question

core identification strategy thus involves recovering preference parameters from respondents' welfare rankings of spending profiles.⁹ Correspondingly, our main survey questions ask respondents for their binary rankings of spending profiles designed to identify parameters of interest. The identifying variations are parameter-specific and are described in Section 3, where the contextual richness facilitates a clearer discussion of the specific identification strategies.

To enhance respondents' intuitive comprehension and facilitate comparisons between spending profiles, visual representations of these profiles were employed (see Figure 1 for two examples). In these graphs, the horizontal axis represents time, with the past on the left, now in the center, and the future on the right. Monthly spending is depicted as bars above the time axis, scaled proportionally to the spending amounts

⁹Because income affects utility through spending, the experiments do not specify the income process except to inform respondents that they can afford the spending profiles in the survey.

and color-coded to distinguish between different temporal periods. For instance, the first spending profile in Figure 1 illustrates monthly spending of \$2,500 in the past up to now, followed by an increase to \$5,000 per month in the future starting now. To ensure accurate interpretation, respondents were provided with detailed instructions and underwent comprehension assessments regarding these graphical representations.

The baseline level of household spending is \$5,000 per month, close to average household spending reported by the Bureau of Labor Statistics and familiar to our respondents, whose average household spending was \$4,362 and \$4,265 in the two survey waves, respectively. Several robustness checks show that deviations between hypothetical and actual spending do not drive the estimates. First, half of the respondents with monthly spending below \$4,000 were randomly assigned a \$3,000 monthly baseline spending; their estimates do not differ significantly from those of the default-baseline group (Column 1 of Table A.4). Second, current spending, expected spending, and the \$3,000-baseline indicator exert no statistically significant impact on the estimates of the preference parameters at the 5% level (Appendix Table A.3)—a result that remains robust across a variety of specifications. Third, restricting the sample to respondents whose actual spending lies close to the baseline spending gives similar results (Column 2 of Table A.4).

The spending profiles presented in our survey involve variations in both the level and timing of spending, designed to mimic common, real-world household spending fluctuations—approximately 50% of U.S. households reported changes in monthly spending from the prior year in 2024 (Larrimore et al., 2025), and checking account data show that the typical median-income U.S. household experiences average month-to-month spending fluctuations of 29% (Farrell and Greig, 2017). In our experiments, the monthly spending varies no faster than the annual frequency, with the average variation kept within 35% of the baseline spending level. To simplify the variations while preserving identification, spending deviates from the baseline only across two stylized time frame categories: (1) all past or all future years and (2) the last or next year. When questions involve other people’s spending, we exclusively use the simpler variation type (1). The levels of spending deviations were chosen to represent meaningful economic changes and were refined through pilot studies.¹⁰ The specific combinations of spending time frames and levels were guided by the identification propositions (e.g., Propositions 1–3), ensuring that the elicitation accurately covers an

¹⁰Pilot studies indicated that respondents found absolute-level framing of spending variations more intuitive and easier to evaluate than growth-rate framing; we therefore use absolute levels throughout.

informative part of the domain for each parameter of interest. Finally, comprehension checks (in module iv) confirm that respondents effectively engaged with the spending variations: 58% of respondents passed the checks on their initial attempt, and the remaining 42% passed on their second attempt.

Hypothetical scenarios involving future spending are routinely used in preference elicitation (e.g., Barsky et al., 1997; Ameriks et al., 2020). Since habit is shaped by the past, the identification of its parameters requires variations in past spending. Our experimental approach of constructing hypothetical scenarios involving past spending, while new in this context, is firmly grounded in counterfactual thinking—a fundamental cognitive process ubiquitous across human reasoning, such as learning, problem-solving, and causal inference (Epstude and Roese, 2008). This approach is exemplified by intuitive reflections such as, “If I had spent less back then, I would be in a better place now,” illustrating how people readily contemplate historical variations. Respondents’ open-ended explanations of their choices show that they can evaluate these counterfactual past variations (Section 3.1). To help distinguish the spending profiles specified in the questions from respondents’ own spending histories, the survey employs a parallel-universe analogy in which all factors are identical except for the spending profiles. Respondents demonstrated a strong grasp of this analogy, with 93% passing the relevant comprehension check on the first attempt. Respondents then choose which universe would yield a more satisfying past or future experience—how they feel about the past up to now or the future starting now. Figure 1 illustrates a typical question.

To reduce cognitive load and control for potential confounding factors, while preserving identification, respondents’ choices are situated within the following hypothetical scenario (with emphasis as in the survey):

To simplify the comparison of your experience under various spending patterns, please evaluate your experience in the following hypothetical situation:

- *There is **no inflation**, and prices of everything stay the same over time.*
- *You **rent the durable goods** you use, including residence, furniture, car, etc.*
- *Your **preferences do not change** over time.*
- ***People not mentioned in the questions always spend \$5,000 per month**, regardless of their income or wealth.*
- ***Any other factors not specified in the questions are identical between the universes and remain***

identical over time.

The hypothetical situation alleviates respondents’ concerns regarding changes in the purchasing power of money, the acquisition of durable goods, and potential shifts in preferences. It is standard practice to exclude inflationary dynamics from hypothetical spending scenarios when inflation is not the primary focus of investigation (e.g., Kapteyn and Teppa, 2003). The acquisition of durable goods can be seen as antithetical to habit formation, as the former tends to increase utility, whereas the latter reduces it. Renting durable goods mitigates this confounding effect. Consistent with the scenario, the literature finds that people typically operate under the assumption of preference stability (e.g., Loewenstein et al., 2003). Because the spending of others may influence individual well-being, the hypothetical situation controls for this factor when its variation is not part of the identifying variation. The scenario holds all other factors constant across universes and over time. To reinforce this point, the instructions clarify that *“The last point means that factors, including but not limited to your income, savings, wealth, control over finances, and the state of the economy, are the same between the universes and remain the same over time. **Only your spending mentioned in the questions varies between the universes**, so please focus solely on those spending differences when making your evaluations.”*¹¹ This ceteris-paribus instruction is a structural feature of preference elicitation that traces back at least to Barsky et al. (1997), whose foundational measurements of preference parameters from hypothetical-choice questions likewise required respondents to hold non-treatment factors implicitly constant; we assess its effectiveness in our setting through the comprehension checks and open-ended explanations documented below. The instructions explicitly define “you” as your household and “others” as other households. Both entities are described as being able to afford the spending specified in the survey questions.

The abstraction in our hypothetical scenarios is a deliberate choice to minimize hypothetical bias and sharpen identification. Adding realistic but irrelevant details, such as inflation dynamics, would raise cognitive burden and reduce engagement, which tends to amplify, rather than reduce, hypothetical bias

¹¹To address potential confusion regarding the relationship between income, saving, and spending, the survey instructions include a clarifying note: *“Even though your income, saving, and wealth are the same across universes, your spending doesn’t have to be identical. For the purposes of this survey, you can imagine a hypothetical external financial system that automatically adjusts your savings at no cost to you, allowing your spending to vary while keeping everything else unchanged.”* This is a deliberate design choice to isolate effects from consumption, free from confounding wealth effects or savings optimization. While this might be confusing for some respondents, we can detect such misunderstandings using open-ended explanations of respondents’ choices, and our results are robust to removing confused respondents (Appendix D.5).

(Haghani et al., 2021b). Leaving confounds uncontrolled risks mixing habit formation with factors such as ownership of durable goods. Our design therefore varies total consumption while holding other factors fixed, matching how people naturally evaluate counterfactuals by changing only the key variable of interest.

To ensure comprehension, key aspects of the hypothetical situation are bolded in the survey (as shown above), and respondents had to pass comprehension tests on both the hypothetical scenario and the monthly spending graphs before reaching modules on preference parameters. For the comprehension checks on the hypothetical situation (module v), 93% of the participants passed on the first attempt and the remaining 7% passed on the second attempt. To assess sustained understanding of the situation, the survey repeated the same checks near the end (in module xi): 92% answered all six questions correctly, 5% missed one, and 3% missed two or three.

The survey was described to respondents as a study of spending behavior, which is an accurate non-specific characterization of the binary-choice task, rather than as a study of habit formation specifically. This framing was chosen for three reasons. First, to mitigate priming bias: revealing the focus on habit formation could undermine the validity of the test for its existence. Second, to prevent confusion, since the average respondent is unfamiliar with habit formation as modeled in economics. Third, to minimize selection on the survey topic. This practice is standard in preference measurement studies, which are routinely described to participants by their broad domain rather than by the specific construct under measurement.

The main survey questions use binary choices to reduce cognitive load. To enhance the precision of the estimates for point-identified parameters, an adaptive unfolding bracket methodology is implemented.¹² This approach entails an initial ranking of two spending profiles, followed by a subsequent ranking of the preferred profile against a third profile that is dynamically adjusted based on the respondent's prior selection. An illustration of this technique is provided in Section 3.2. The binary-choice format deliberately excludes an option for indifference, thereby reducing the propensity for satisficing behavior (Caplin et al., 2011) and fostering more thorough engagement with the presented scenarios.¹³ When respondents

¹²This design diverged from Ameriks et al.'s (2020) strategic survey question format because the identifying variations encompass both spending levels and time horizons. This multifaceted structure raises the cognitive burden, rendering their approach impractical for eliciting indifference points here.

¹³The sole exception pertains to the time discount rate question regarding past experiences, where the spending profiles across options were objectively identical, and participants were offered the opportunity to indicate indifference.

perceive the options as equivalent, their selections are presumed to be stochastic, consistent with stochastic choice theory (Block and Marschak, 1960).

The main survey questions elicit preferences rather than objectively correct or incorrect answers, so direct incentivization is infeasible. To counteract potential biases arising from misinterpretation, inattention, or protest responses, the survey instrument was designed to minimize cognitive load: visual spending profiles, repeated emphasis that spending is the only difference between response options, explicit clarification of time horizon with spending differences, and follow-up questions with dynamic explanations of spending changes. To facilitate the distinction between past and future experiences, survey questions solicited responses for both, with definitions reiterated and the words “past” and “future” highlighted in each question. Interactive answer options darken when hovered over or selected, helping respondents avoid accidental clicks.

To ensure sustained participant attention throughout the survey, a layered approach incorporating both explicit and implicit attention checks was implemented. Explicit checks consist of initial comprehension questions and targeted questions with specific expected responses. Implicit checks include response time, exit comprehension checks, and cross-wave response consistency. Participants knew that attention checks existed but not where they appeared. To further incentivize diligent participation, high-quality responses entered into a lottery with a 1 in 100 probability of winning \$5.¹⁴ The survey also asked respondents to self-report attention and effort: over 99% of participants reported maintaining full attention midway through the survey. At the end, 82% reported maximum effort, 16% substantial effort, and 1% minimal effort.

To further enhance reliability, we follow Barsky et al. (1997) and conduct the survey in two waves, with the first wave fielded in August 2024 and the second wave two weeks later in September 2024. In the second wave, the core modules were reordered, and the order of response options was reversed to mitigate order-related response biases. To reduce respondent burden, the second wave omitted the demographics module and the two modules of calibration questions. The baseline spending level from the first wave was retained in the second wave. The test-retest correlations for the core module responses averaged 0.45 (see Appendix Table A.1 for parameter-specific correlations across the waves),¹⁵ which falls

¹⁴From the 2,097 valid responses, 21 respondents were randomly selected for this award.

¹⁵The test-retest correlation for the responses on external habit formation was excluded from this calculation as the absence

within the typical reliability range (0.15–0.60) for preference measurements through choice experiments (Chuang and Schechter, 2015). Estimates using only first-wave data (Column 5 of Appendix Table A.4) are quantitatively close to the benchmark estimates, indicating that the second wave serves to mitigate response error and to enable the test-retest validity check, rather than to drive the estimates.

It is worth emphasizing that comparisons of the spending profiles do not amplify the significance of habit formation. Comparative evaluations do not intrinsically result in estimates indicative of habit formation ($u_H < 0$); these evaluations are equally compatible with $u_H \geq 0$, under which habit formation is not present. Importantly, neither the experimental design nor the estimation procedure constrains the sign of u_H , thereby allowing the data to reveal the genuine effect of habit formation without introducing bias. Whereas real-world behavior is frequently subject to many intersecting factors that obscure the influence of habit formation, the controlled experimental environment is deliberately structured to isolate consumption patterns needed to identify key habit parameters, thus providing evidence that is difficult to obtain from real-world contexts.

2.3 Validity of Hypothetical Choices

Systematic deviations between hypothetical and real-world choices are typically categorized as hypothetical bias (Haghani et al., 2021a). Our design therefore targets the most likely sources of hypothetical bias in the context of eliciting preferences over total-consumption profiles.

The main threats in this setting are cognitive bias and lack of consequentiality (Haghani et al., 2021a; Johnston et al., 2017). Cognitive bias is the sharper concern: respondents in an unemotional “cold” state may mispredict utility under different consumption profiles. To mitigate this concern, our experimental design anchors the scenarios to empirical realities, a strategy known to enhance the validity of elicited responses (Manski, 2004; Haghani et al., 2021b). Specifically, the baseline monthly household spending level of \$5,000 closely approximates average U.S. household expenditures reported by the Bureau of Labor Statistics, with 85.9% of respondents’ household spending lying within one standard deviation from this baseline. For respondents with lower actual spending, we randomized half of them into a \$3,000 baseline

of evidence on external habit formation (Section 3.1) suggests that their responses are random and thus naturally have low cross-wave correlation.

spending and find that the estimates from this group are statistically indistinguishable from those of the main sample (Section D.1), indicating that the default baseline spending level does not bias responses. The variations in our consumption profiles also mirror the common life experiences of U.S. residents (Section 2.2). Because consumption choices are familiar, respondents can draw on well-formed preferences, which reduces hypothetical bias (List, 2001). Their open-ended responses also show effective counterfactual reasoning about past spending variations (Section 3.2).

Crucially, our approach does not require respondents to report absolute utility levels. Our method remains valid provided that cognitive biases do not alter the ranking of options—an assumption that holds if biases affect all predictions symmetrically. This distinction is well supported by the literature, which suggests that hypothetical bias tends to distort absolute levels while preserving the reliability of relative trade-offs (see Haghani et al., 2021a, for a review).¹⁶ Since choices are not financially binding, respondents may exert insufficient effort. We address this through several mechanisms detailed below, including comprehension checks and incentivizing response quality.

Other common sources of hypothetical bias, such as social desirability or strategic behaviors, are less pertinent here. Unlike surveys on prosocial or environmental goods, preferences for total-consumption profiles are not laden with strong normative motivations. The risk of respondents trying to “appear good” is therefore minimal. The academic nature of our survey, which was not framed as a commercial or policy pricing study, makes it unlikely that respondents would strategically misrepresent their preferences to influence a market or policy outcome.

To counter these potential biases, we built a suite of ex-ante mitigation strategies into the survey instrument, focusing on contextual realism, cognitive load reduction, and mitigation of inattention and

¹⁶Projection bias (Loewenstein et al., 2003)—the tendency to project one’s current circumstances onto the evaluation of counterfactual conditions—is one example of such a bias: consistent with the distinction drawn above, it distorts the perceived *level* of utility rather than the *ranking* of the universes, so even though it is directional it does not by itself reverse the sign of u_H . Our existence results are therefore robust to it, and at most the estimated *magnitude* of habit decay could be affected. That direction is itself unsettled in a hypothetical, retrospective setting: the projection-bias literature studies *prospective* forecasting, and while a projection or anchoring channel would make our estimate a lower bound on the true decay rate (widening the discrepancy with the asset-pricing–calibrated dynamics in Section 4.1), memory-based channels such as fading-affect bias (Walker et al., 1997) would push toward an upper bound, so we do not rest the magnitude on a presumed direction. Reassuringly, projection bias operates through the gap between a respondent’s actual circumstances and the hypothetical scenario; the \$3,000-baseline arm narrows that gap for lower-spending respondents—leaving less room for current-state projection—yet yields estimates statistically indistinguishable from the main sample.

indifference. By incorporating clear visual representations of spending profiles, a “parallel-universe” analogy, and baselines tied to average consumer expenditures, we mitigate the “hot-cold empathy gap” and limited familiarity, creating a tangible decision-making environment to strengthen internal and external validity. To minimize reliance on simplifying heuristics, we employed a streamlined binary-choice format, provided detailed instructions with comprehension checks, and controlled for complicating factors such as inflation, thereby reducing cognitive burden and encouraging deliberate responses. Additionally, explicit and implicit attention checks, a response-quality lottery, and no indifference option discourage inattention and satisficing. The two-wave design separates preferences from response error. Appendix D shows the robustness of our estimates across demographics, spending baselines, time horizons, response biases and errors with nonzero and wave-varying means, first-wave-only estimation, strict compliance with the ceteris-paribus structure of the hypothetical situation, and additional attention checks.

Given the lack of readily observable counterparts for these structural habit parameters in conventional data, we assess the validity of our hypothetical choices by demonstrating their convergent and predictive validity (Strauss and Smith, 2009). To establish convergent validity, we show consistency of our estimates with findings on related behaviors in the literature: the estimated rate of habit decay closely aligns with prior estimates of income adaptation speed (Section 3.2), and the derived welfare impacts of habit formation and peer effects effectively match established estimates for income adaptation and peer effects, respectively (Section 3.3). Qualitative responses provide further support: about 80% of respondents whose choices indicate internal habit formation cited reasons reflecting the difficulty of cutting back from accustomed spending (Section 3.1). To demonstrate predictive validity, we show that the habit decay rate measured in our survey predicts real-world happiness responses: respondents who reported faster habit adjustment in our hypothetical scenarios also exhibited faster hedonic adaptation to the 2024 U.S. presidential election outcome, which occurred two months after our survey (Section 3.2).

2.4 Data Summary

The survey was administered via Prolific, an online platform designed for human intelligence tasks. To mitigate cultural bias, the sample was restricted to residents of the United States. While online samples may not achieve the representativeness of census-based samples, they offer greater representativeness than in-

person convenience samples (Berinsky et al., 2012). These samples are commonly employed in economic research (e.g., Kuziemko et al., 2015; Bordalo et al., 2016; Kumar et al., 2023; Sergeyev et al., 2024). Beyond this, research indicates that Prolific yields higher response quality relative to other online platforms (Peer et al., 2022; Douglas et al., 2023). Median participants spent about 31 and 23 minutes on the two-wave surveys, earning \$4.70 and \$2.70, respectively, for an effective hourly rate of approximately \$8.00.

This study used Prolific’s representative-sample feature to recruit a sample of U.S. residents. Of the 1,367 completed first-wave responses, 1,243 respondents indicated willingness to participate in a follow-up, and 730 (approximately 59%) completed the second wave. Completion of the survey required passing the in-survey comprehension and attention checks, so responses that failed these checks (or were abandoned mid-survey) are not in the sample by construction. The benchmark estimation includes all such completed responses from both waves.

Appendix Table A.2 presents the sample’s demographic distributions alongside those of the 2020 U.S. Census. The sample is broadly representative on gender, race, household size, and region, but, as is common in online samples, older individuals (65+), those with less than some college education, and those with incomes exceeding \$120,000 are represented differently than in the Census. Appendix D.1 addresses these differences by allowing the estimated preference parameters to vary with observed demographics; none of the demographics is statistically significant after adjusting for multiple hypothesis testing.

2.5 Statistical Model

This study employs a statistical model to extract consistent variations across the two waves of responses and to mitigate response errors not addressed by the survey experimental design. Potential remaining response biases and errors are addressed through robustness checks in Appendix D.

The observed response for preference parameter x can vary across respondents, indexed by i , and waves, indexed by w , and is generated by $X_{i,w} \equiv \sum_{k=1}^K k \cdot 1(T_{k,\tilde{x}} < \tilde{x}_{i,w} \leq T_{k+1,\tilde{x}})$, where k indexes response options and $1(\cdot)$ is the indicator function. The latent variable $\tilde{x}_{i,w} = x_i + \varepsilon_{i,x,w}$, and $T_{\{k\},\tilde{x}}$ denotes the sequence of known thresholds determined by the identifying propositions and the experimental variations (see Section 3 for details), with $T_{1,\tilde{x}} = -\infty$ and $T_{K+1,\tilde{x}} = +\infty$. The true parameter value for individual i , x_i , is drawn from $\mathcal{N}(\mu_x, \sigma_x^2)$. The response error, $\varepsilon_{i,x,w}$, can vary across respondents, parameters, and waves,

and is drawn from $\mathcal{N}(0, \sigma_{\varepsilon_x}^2)$ independently of the true parameter value for each wave. A robustness check allows the means of the response errors to be nonzero and to vary across waves (Appendix D.3), finding that the estimated means are indistinguishable from zero and that the estimates of the preference parameters are not significantly different from those under the specification here. For aggregation and computation, the parameters are assumed to be independent within a respondent. Because the respondents are geographically dispersed across the United States and are unlikely to be acquainted, the responses are assumed to be independent across respondents. Allowing the response errors to persist across waves (i.e., $\sigma_{\varepsilon_x, P}^2 = \text{Cov}(\varepsilon_{i,x,1}, \varepsilon_{i,x,2})$ and $\sigma_{\varepsilon_x}^2 = \sigma_{\varepsilon_x, P}^2 + \sigma_{\varepsilon_x, T}^2$, where P and T indicate the persistent and transitory components, respectively), we arrive at the joint distribution of respondent i 's parameter x in the two waves of the survey:

$$\begin{bmatrix} \tilde{x}_{i,1} \\ \tilde{x}_{i,2} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mu_x \\ \mu_x \end{bmatrix}, \begin{bmatrix} \sigma_x^2 + \sigma_{\varepsilon_x, P}^2 + \sigma_{\varepsilon_x, T}^2 & \sigma_x^2 + \sigma_{\varepsilon_x, P}^2 \\ \sigma_x^2 + \sigma_{\varepsilon_x, P}^2 & \sigma_x^2 + \sigma_{\varepsilon_x, P}^2 + \sigma_{\varepsilon_x, T}^2 \end{bmatrix} \right),$$

whose cumulative distribution function is denoted by Φ .

Given the predominance of representative-agent frameworks in contemporary habit formation models, this study examines the implications of parameter estimates within such models. In Appendix B, we establish that individual agents' parameter values aggregate to their population mean for the representative agent. Formally, let x_R denote the representative agent's value for parameter x , and let x_i represent the corresponding parameter value for individual i . We show that $x_R = \frac{1}{N} \sum_{i=1}^N x_i$, where N is the number of individuals. Since $\frac{1}{N} \sum_{i=1}^N x_i \xrightarrow{p} \mu_x$, the parameter of interest in estimation is μ_x , the mean of the individual parameter values.¹⁷

Estimates are derived from joint maximum likelihood estimation, with the likelihood function

$$\prod_{i,x} \left[\Phi \left(\begin{bmatrix} T_{X_{i,1}+1, \tilde{x}} \\ T_{X_{i,2}+1, \tilde{x}} \end{bmatrix} \right) - \Phi \left(\begin{bmatrix} T_{X_{i,1}+1, \tilde{x}} \\ T_{X_{i,2}, \tilde{x}} \end{bmatrix} \right) - \Phi \left(\begin{bmatrix} T_{X_{i,1}, \tilde{x}} \\ T_{X_{i,2}+1, \tilde{x}} \end{bmatrix} \right) + \Phi \left(\begin{bmatrix} T_{X_{i,1}, \tilde{x}} \\ T_{X_{i,2}, \tilde{x}} \end{bmatrix} \right) \right].$$

For respondents observed only in the first wave, the second-wave response is treated as missing and its probability contribution is set to one, so their likelihood contribution reduces to the corresponding

¹⁷ σ_x^2 is not separately identified, since the cross-wave covariance pins down only $\sigma_x^2 + \sigma_{\varepsilon_x, P}^2$; this does not affect identification of the parameter of interest μ_x .

first-wave marginal probability. To address the computational challenges posed by joint estimation, we adopt a Bayesian approach, thereby circumventing the need for direct maximization of the likelihood function. Specifically, we employ Hamiltonian Monte Carlo, a Markov Chain Monte Carlo method known for its superior sampling efficiency, together with uniform priors for the main parameters of interest to ensure equivalence between posterior modes and maximum likelihood estimates.

3 Measurement

This section details the measurement of the preference parameters of interest, including the identifying variations, survey questions, and resulting estimates.

3.1 Existence of Habit Formation

Existence of internal habit formation. Under the functional $u(C, H)$, habit formation is the property that a higher habitual level of spending lowers current utility; that is, $u_H < 0$. Because habit, H , increases with past consumption, internal habit formation is consistent with the utility differential $Q_{HI} \equiv u(C, H + \Delta h) - u(C, H)$ being negative for $\Delta h > 0$ stemming from higher past own consumption, but is inconsistent with a non-negative differential.

To identify the sign of Q_{HI} , it is thus intuitive to vary the respondent's past spending while controlling for future spending, so that variation in future experience comes only from variation in habit. Figure 1 shows a screenshot of the resulting survey question eliciting the sign of Q_{HI} .¹⁸ In this question, preferring a spending profile with low past spending (low H) over one with high past spending (high H) for a better future experience implies $Q_{HI} < 0$. In other words, choosing Universe One in the question shown in Figure 1 is consistent with internal habit formation, while choosing Universe Two is not. It is worth emphasizing again that the survey does not prime respondents with habit formation and that no assumption is made about the sign of Q_{HI} . Additionally, comparison (over time) does not necessarily imply $Q_{HI} < 0$, because the comparison could also be consistent with $Q_{HI} \geq 0$, in which case internal habit formation would not exist.

Response distributions to this question show that about two-thirds of respondents chose lower past

¹⁸In Figure 1, past spending deviates by \$2,500 below or above the baseline spending of \$5,000. In the survey version with the baseline spending of \$3,000, past spending varies by \$1,000 below or above the baseline.

Table 1: Response Distributions

	First wave		Second wave	
Panel A: Sign-identified parameters	U1	U2	U1	U2
Existence of internal habit formation	64.3%	35.7%	65.3%	34.7%
Existence of external habit formation	51.1	48.9	53.2	46.8
Panel B: Point-identified parameters	U1U1	U1U2	U2U1	U2U2
Habit decay rate	34.7%	19.5%	18.5%	27.4%
$-u_H/u_C$	34.1	8.6	16.9	40.5
$u_{C_{\text{others}}}/u_H$	48.1	14.0	6.8	31.0

Notes: Sample: 2,097 observations. Entries are percentages. UX: Universe X. U1U2: chose Universe One in the initial question and Universe Two in the follow-up; analogous notation applies to other response sequences.

spending for a better future experience—Universe One—in each wave of the survey (Panel A of Table 1), consistent with the existence of internal habit formation. After accounting for response errors, the estimate of $\text{sgn}(Q_{HI})$ corroborates this finding (Table 2).

At the conclusion of the survey, respondents provided open-ended explanations of how they arrived at their answers to the question regarding the existence of internal habit formation.¹⁹ The full question and their choices were displayed. Using machine learning with human verification, we categorize the explanations and rank them by frequency. Table 3 presents the most common explanations.

Since the precise concept of habit formation typically lies outside conscious awareness (Wood and R  nger, 2016), respondents might not accurately attribute their choices to it. Therefore, we interpret responses such as “increased (future) spending (relative to past) leads to (more) satisfaction” as evidence that habit formation influenced their decisions. For instance, one respondent noted, “It’s the same amount of money, but because I spent way more in the past in Universe Two than One, the amount in One would FEEL like more than it’s in Two. Spending more is never a problem; it’s having to cut down from spending that’s hard” (respondent’s emphasis). Another stated, “Because I’m accustomed to spending less (\$2,000) a month, if I have more money to spend, my future will be happier and more satisfying.” As shown in Table 3, about 80% of respondents who selected Universe One ($Q_{HI} < 0$) cited reasons consistent with habit formation, while the remaining 20% indicated misunderstanding of the hypothetical scenario (e.g.,

¹⁹To maintain a reasonable survey duration, respondents were not asked to explain their responses to other questions. For those questions, we assess validity through the shared comprehension checks, along with the convergent and predictive validity checks where possible.

Table 2: Parameter Estimates

	Estimate	95% CI
$\text{sgn}(Q_{HI})$	-1	{-1}
$\text{sgn}(Q_{HE})$	1	{-1, 0, 1}
Habit decay rate	1.03	[0.86, 1.17]
Habit decay factor (annual)	0.64	[0.58, 0.69]
$-u_H/u_C$	0.63	[0.57, 0.71]
$u_{C_{\text{others}}}/u_H$	0.30	[0.13, 0.47]
$u_{C_{\text{others}}}/u_C$	-0.19	[-0.30, -0.08]
$u_H/u_C + u_{C_{\text{others}}}/u_C$	-0.82	[-0.96, -0.69]

Notes: Annual habit decay factor: $1 - e^{-\text{Habit decay rate}}$. $u_{C_{\text{others}}}/u_C$ and $u_H/u_C + u_{C_{\text{others}}}/u_C$ are implied by $-u_H/u_C$ and $u_{C_{\text{others}}}/u_H$. $\text{sgn}(\cdot)$ takes values in $\{-1, 0, +1\}$, and its 95% CI denotes the set of sign values not rejected at the 5% level.

incorrectly assuming higher income or savings in Universe One) or general errors.

The reasoning of respondents who selected the option inconsistent with habit formation (Universe Two, $Q_{HI} > 0$) also provides insight. Approximately two-thirds of these respondents cited a preference for “decreased (future) spending (relative to past),” often associating it with frugality, better financial control, or greater efficiency. While these reasons suggest an implicit misunderstanding of the hypothetical scenario—in which unspecified factors such as frugality, financial control, and efficiency remain constant across time and response options—they do not contradict habit formation. Other respondents chose Universe Two due to more explicit misunderstandings, such as ignoring the survey instruction to rent durable goods, misinterpreting the constancy of income and savings, or making general errors or random choices. These responses highlight a practical trade-off in survey design: more complete ceteris-paribus instructions can reduce misunderstandings but make the scenario longer and harder to process. They also suggest that responses inconsistent with internal habit formation largely reflect added considerations or misunderstandings, rather than reasoning that contradicts internal habit formation itself.

Chetty and Szeidl (2016) propose that consumption commitments, arising from the costly adjustment of illiquid durable goods such as residences and vehicles, provide an alternative explanation for consumption patterns typically ascribed to habit formation. To distinguish between these concepts, this study employs a hypothetical scenario in which respondents rent durable goods to minimize adjustment costs. It includes only responses from participants who demonstrate a clear understanding of the scenario in

Table 3: Rationale and Proportion of Respondents' Choices

Rationale	Proportion		
	First wave	Second wave	Overall
Choose U1 ($Q_{HI} < 0$) because			
Increased spending leads to satisfaction	76.0%	81.5%	50.3%
More saving	8.0	5.0	4.5
More income in the future	5.5	7.8	4.1
Other reasons	10.6	5.7	5.7
Total	100.0	100.0	64.6
Choose U2 ($Q_{HI} > 0$) because			
Reduced spending leads to satisfaction	63.9	71.5	23.5
Durable spending	18.5	13.4	5.9
Increased saving in the future	10.9	6.7	3.3
Other reasons	6.8	8.3	2.6
Total	100.0	100.0	35.3

Notes: Primary rationales for choices in the internal-habit existence question. Entries are percentages.

the comprehension checks, with approximately 94% of them also indicating such understanding in the open-ended feedback discussed above. The survey question uses symmetric deviations between past and future spending, thereby excluding residual psychological or non-psychological factors, captured by symmetric (e.g., quadratic) adjustment costs, as drivers of the observed responses. The open-ended feedback also indicates that the vast majority of respondents' choices are not driven by costs related to adjusting consumption commitments. These findings suggest that habit formation can influence behavior independently of consumption commitments in scenarios that minimize adjustment costs.

Existence of external habit formation. External habit formation is consistent with lower future utility resulting from higher past spending by others. Allowing H to increase with others' past consumption, external habit formation predicts that the utility differential $Q_{HE} \equiv u(C, H + \Delta h) - u(C, H)$ is negative for $\Delta h > 0$ arising from higher past spending by others, but is inconsistent with a non-negative differential.

To identify the sign of Q_{HE} , it is natural to vary other people's past spending while holding constant both their future spending and the individual's own past and future spending. This ensures that variation in future experience is driven solely by changes in habit due to other people's past spending. To visually represent these variations, the spending graphs were augmented with others' spending profiles. To facilitate comparisons over time and across people, one's own spending graph is compressed horizontally so that both one's

Remember, future experience reflects how you feel about the future starting now.
Which universe will provide **you** with a more satisfying **FUTURE** experience?



Figure 2: Survey Question on Existence of External Habit Formation

own and others' spending graphs fit on a single line (see Figure 2). Different colors distinguish between the spending of oneself and others. Respondents received a brief instruction about these visual changes and were informed that the compressed graphs encompassed the same time frames as the uncompressed graphs.

In this context, preferring a spending profile with low past spending by others (low H) over one with high past spending by others (high H) implies $Q_{HE} < 0$. In other words, choosing Universe Two in the question is consistent with external habit formation, while choosing Universe One is not. Since the survey does not prime respondents with habit formation and no assumptions are made about the signs of derivatives of the utility function with respect to habit, comparison (over time and across people) does not necessarily imply $Q_{HE} < 0$ (that external habit formation exists), because the comparison could also be consistent with $Q_{HE} \geq 0$, in which case external habit formation does not exist.

The distribution of responses (Panel A of Table 1) suggests that respondents essentially choose randomly between the two universes in the question shown in Figure 2, providing no evidence that others' consumption systematically influences one's habit. After accounting for response errors, the estimate of $\text{sgn}(Q_{HE})$ includes 0 in its 95% confidence interval (Table 2), indicating no evidence for external habit formation.

As discussed in Section 2.2, the survey questions exclude an indifference response option. The essentially uniform distribution of responses to this question supports this design choice, confirming that indifferent respondents tend to choose randomly in the absence of such an option. The finding of no evidence for external habit formation aligns with the evidence from Brunnermeier and Nagel (2008) and, combined with our evidence on internal habit formation, is consistent with models featuring internal habit formation while challenging those relying solely on external habit formation.

Alternative interpretations of the responses. Throughout the paper, we define and measure habit formation at the level of the functional $u(C, H)$, as formalized in Section 2.1. Mechanisms that generate the same functional are observationally equivalent and equally suitable for our applications. The remaining alternatives fall into two groups: preference theories that generate predictions differing from those of $u(C, H)$ within our design, and the possibility that responses reflect features of the elicitation rather than preferences of any kind. Although our design was not constructed as a horse race among mechanisms, each of these alternatives carries implications for response patterns that the survey already reports, and the reported patterns are difficult to reconcile with them.

Consider first reference dependence with expectations-based reference points (Kőszegi and Rabin, 2006). In our design, each universe's full spending path is known and anticipated, so realized consumption coincides with its expectation in every period of both universes. Utility is then identical across the two universes, and the model predicts an even split in the question of Figure 1 rather than the roughly two-thirds majority observed in each wave. Variants in which the reference point adjusts toward lagged beliefs become backward-looking and rejoin the equivalence class of Section 2.1, as do status-quo reference points (Masatlioglu and Ok, 2014), which constitute its fast-adjustment limit.

A second alternative holds that people have genuine preferences over properties of consumption sequences, such as improvement over time (Loewenstein and Sicherman, 1991; Loewenstein and Prelec,

1993). To the extent that such preferences operate by comparing consumption with the accustomed past, their backward-looking component again belongs to the equivalence class; their distinctive, non-nested content is a taste for improvement within the prospective profile. The time-discount question (Appendix E.1) bears on this content: it contrasts a transitory near-term increase that subsequently returns to baseline against a smaller permanent increase, and a taste for improving prospective sequences would favor the permanent increase, which contains no later decline. Respondents chose the transitory profile often enough to yield an estimated discount rate of $\rho = 0.10$ (standard error 0.01), within the conventional range of impatience (Matousek et al., 2022), with no sign that a dislike of the later decline systematically tilts choices toward the permanent profile. Because the transitory profile is also the larger near-term increase, however, this choice trades off impatience against that dislike rather than isolating it. The evidence is therefore directional—indicating that any prospective taste for improvement is not strong enough to overturn conventional impatience at these magnitudes—rather than a clean rejection of such a taste, beyond what backward-looking dependence on past consumption already captures.

A separate concern is that responses reflect not preferences at all, but a graphical heuristic: choose the upward-sloping spending profile. The external-habit question provides a placebo for this account. It varies the slope of others' spending while holding the respondent's own spending fixed. A purely visual preference for rising profiles would therefore lead respondents to choose the universe in which others' spending increases. Responses instead split nearly evenly, 51/49 and 53/47 across the two waves (Table 1), indicating that a rising graphic alone does not drive choices. This placebo is not a perfect test: because it varies others' rather than the respondent's own profile, it addresses a label-insensitive attraction to any rising graphic, not a narrower heuristic tied to one's own rising profile. The open-ended rationales discussed above speak to that narrower account: about 80% of respondents choosing consistently with internal habit cite the difficulty of adjusting to an unaccustomed level of own spending rather than the shape of the graph.

Finally, memory-based channels warrant comment because they can enter utility in either direction. Memory-mediated habituation, under which higher past consumption lowers the current evaluation of a given spending level, implies $u_H < 0$ and is itself a member of the equivalence class above rather than a competing alternative. The one memory channel with the opposite sign is savoring—deriving

present pleasure from recollected high past consumption—which implies $u_H > 0$ and would make the high-past-spending universe the more satisfying one going forward. Both the estimated $u_H < 0$ and respondents’ open-ended explanations run against this savoring channel.

3.2 Speed of Habit Decay

The rate of habit decay is governed by θ , as in $\dot{H} = \theta (C - H)$. A natural survey question to elicit θ therefore varies the persistence of past spending. To point-identify θ , the level of past spending must also vary to ensure a surjective mapping from θ to survey responses. Inverting this mapping recovers values of θ consistent with the responses, as formalized in Proposition 1. Appendix A provides proofs for all propositions.

Proposition 1. $\theta > -\ln \left(1 - \frac{\Delta C_{U1}}{\Delta C_{U2}} \right)$ if a respondent chooses Universe One over Universe Two for a more satisfying future experience in a habit decay rate question.²⁰

Figure 3 presents a screenshot of the survey question for identifying θ . The question asks respondents to choose the spending profile that would yield a more satisfying future experience: either a medium-high level of spending spread over an extended period in the past (Universe One), or a very high level of spending concentrated in the recent past (Universe Two), while future spending remains constant. The intuition behind this identification strategy is that a preference for the profile with more persistent past spending (Universe One) implies that recent past spending contributes more to habit than distant past spending. This suggests that habit decays quickly, relative to the threshold $-\ln \left(1 - \frac{\Delta C_{U1}}{\Delta C_{U2}} \right)$, as specified in the proposition.

ΔC_{U1} and ΔC_{U2} denote the deviations in the levels of monthly spending from the baseline spending level in Universe One and Universe Two, respectively. In Figure 3, $\Delta C_{U1} = \$2,000$ and $\Delta C_{U2} = \$4,000$. Thus, the survey question in Figure 3 partitions the possible values of θ into two complementary intervals, separated by the threshold $\ln 2$, which corresponds to an annual habit decay speed of 50%. Respondents selecting Universe One indicate a habit decay speed exceeding 50% per year, while those choosing Universe Two reveal a decay speed below this threshold.

²⁰The paper’s identifying propositions are stated as conditional statements, sufficient for its results, though all can be strengthened to biconditional (if and only if) statements. The notation of ΔC_{U1} and ΔC_{U2} is defined below.

Remember, future experience reflects how you feel about the future starting now.
Which universe will provide you with a more satisfying **FUTURE** experience?

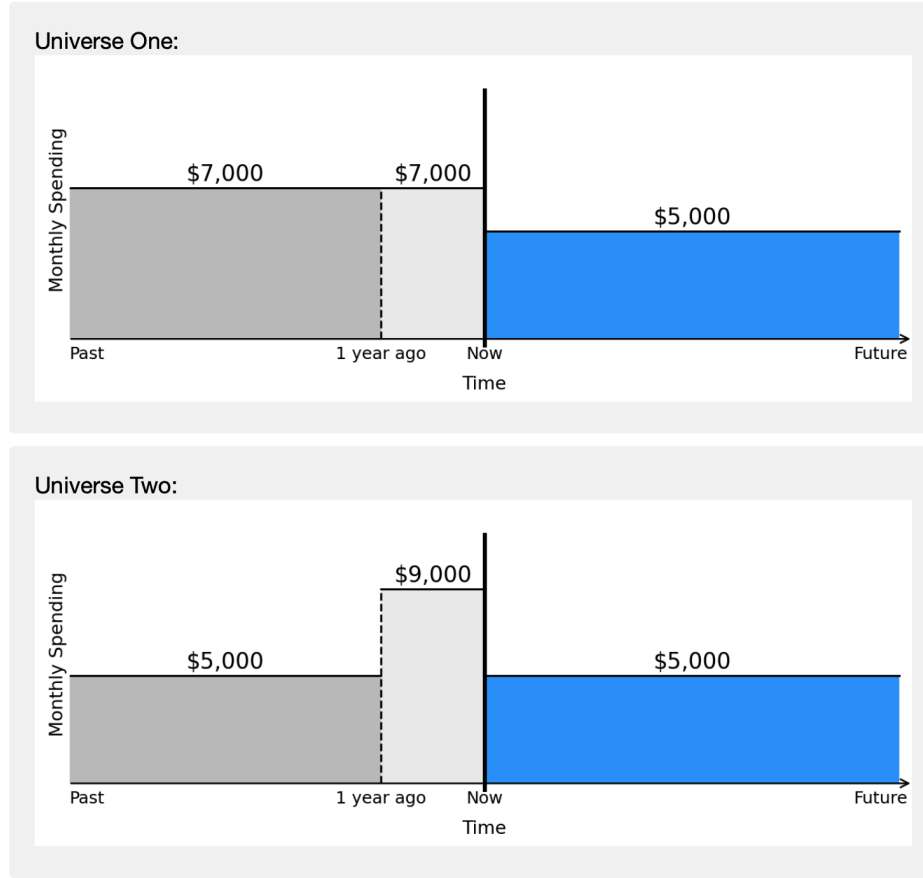


Figure 3: Survey Question on Speed of Habit Decay

To improve accuracy, the survey uses unfolding brackets to sequentially narrow the range of θ values based on respondents' choices. Each respondent answers a follow-up question that places their θ within one of four predefined brackets (see Figure 4). These brackets are designed to ensure that the spending profiles are easily comprehensible to respondents and that they cover an informative range of the parameter space, with the design refined through pilot studies. If a respondent selects Universe One (U1) in the initial question, indicating that their (annualized) θ exceeds 0.5, they are then presented with a follow-up question tied to a threshold of 0.9 (see Appendix Figure A.2 for a screenshot of the question). If they subsequently choose Universe Two (U2), the module concludes, indicating that their habit decay factor lies between 0.5 and 0.9. Table 4 details all thresholds and the corresponding values of ΔC_{U1} and ΔC_{U2} used in the survey. Because the first and last brackets are open-ended, nontrivial mass in boundary brackets

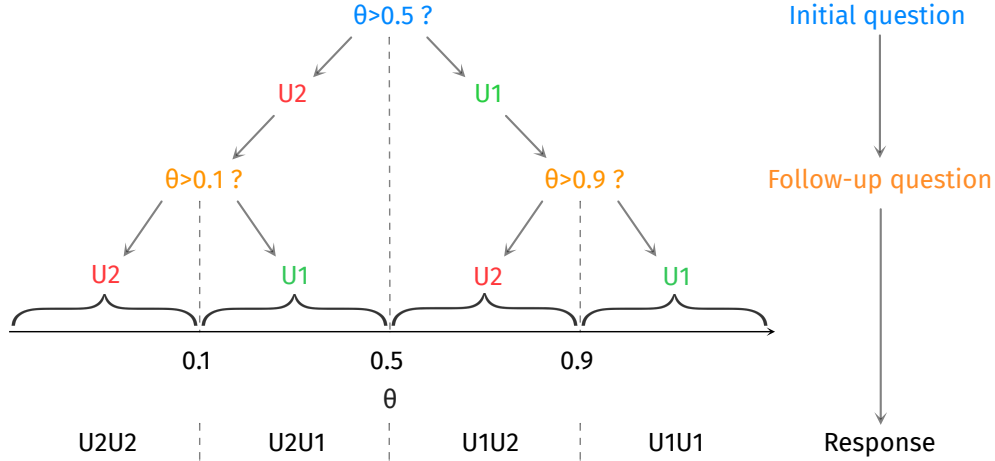


Figure 4: Unfolding Brackets

Notes: U1 and U2: Universe One and Universe Two, respectively. Habit decay rates annualized.

is not by itself evidence of bimodality in latent preferences; such mass can also arise from response error. Estimation using the two-wave data helps separate preferences from response error.

Applying the statistical model described in Section 2.5 to the responses yields an estimate of 1.03 for the habit decay rate, corresponding to an (annual) habit decay factor of 0.64 (Table 2). This decay speed implies that nearly 90% of habit depends on spending over the last two years, which is close to findings in the psychology literature indicating that income adaptation takes about two years (for a review, see Clark et al. 2008). The estimate's narrow confidence interval ([0.58, 0.69] at the 95% level), especially relative to the range used in habit-based models, suggests that calibrated decay rates often reflect model-fitting targets more than direct evidence on the persistence of habit. Section 4.1 illustrates this point in the canonical habit formation model of Campbell and Cochrane (1999).

Predicting hedonic adaptation speed. As a validation of our measure of habit decay and our broader methodological approach, we link responses from our survey to responses from a separate survey examining happiness reactions to the 2024 U.S. presidential election (Kimball et al., 2024a).²¹ The happiness survey, conducted independently in November and December 2024 on Prolific, included approximately 9,400 respondents. By matching anonymized unique identifiers, we identify 503 of their respondents who also participated in our survey.

²¹ Respondents consented to the potential linkage of their data across multiple surveys.

Table 4: Quantities and Thresholds for Spending Profiles

Question	Quantity and threshold	Baseline = \$5,000			Baseline = \$3,000		
		Choosing U2	Initial question	Choosing U1	Choosing U2	Initial question	Choosing U1
Habit decay rate	ΔC_{U1}	400	2000	2000	200	1000	1000
	ΔC_{U2}	4000	4000	2200	2000	2000	1100
	Threshold	0.11	0.69	2.40	0.11	0.69	2.40
$-u_H/u_C$	ΔC_{past}	2000	2000	2000	1000	1000	1000
	ΔC_{future}	1000	200	20	500	100	10
	Threshold	0.92	0.55	0.10	0.92	0.55	0.10
$u_{C_{\text{others}}}/u_H$	ΔC_{others}	200	200	1000	100	100	500
	ΔC	4000	2000	2000	2000	1000	1000
	Threshold	1.80	0.90	0.18	1.80	0.90	0.18

Notes: Quantities and implied thresholds mentioned in Propositions 1–3. “Choosing U1” and “Choosing U2”: follow-up questions shown after the initial Universe One or Universe Two choice. Quantities are in U.S. dollars.

The happiness survey measures respondents’ happiness both before and after the 2024 U.S. presidential election using a standardized question: “On a scale of 0 to 100, where 0 is the lowest level possible and 100 is the highest level possible, how happy do you feel right now?” Supporters of the election winner tend to report higher happiness, while supporters of the loser tend to report lower happiness post-election, with both groups exhibiting hedonic adaptation toward their pre-election happiness levels (Kimball et al., 2024a). We use the survey’s post-election waves to construct a measure of the speed of hedonic adaptation to the election outcome. Specifically, we calculate the absolute daily change in happiness for each respondent between two post-election survey waves at times t_1 and t_2 : $\left| \frac{Happiness_{t_2} - Happiness_{t_1}}{t_2 - t_1} \right|$. The median date for t_1 is November 6, 2024, coinciding with the announcement of the election results. To capture the rapid nature of hedonic adaptation to election news, the time difference between t_2 and t_1 is randomized, ranging from 1 to 19 days, with a median of 4 days.

In the linked dataset, we find a statistically significant correlation (0.13, p-value 0.004) between responses to our survey question on habit decay speed and the hedonic adaptation speed. This suggests that individuals with faster habit decay and thus faster habit adjustment, as measured by our approach, exhibit faster happiness adaptation to the election outcome. Although consumption habit formation and hedonic adaptation are different, their plausible connection and the potential nonlinearity of this relationship render the correlation of 0.13 economically meaningful.

Table 5: Habit Decay Speed Predicts Hedonic Adaptation Speed

	Absolute daily happiness changes	
	(1)	(2)
Fast habit decay (>50%/year)	1.31 (0.48)	1.30 (0.49)
Constant	3.23 (0.35)	5.39 (3.76)
Demographics	N	Y
Baseline spending levels	N	Y
Observations	503	503

Notes: Regression results of absolute daily happiness change on the indicator for annual habit decay speed above 50%. Column 2 adds controls for age, age squared, gender, household income, historical and expected monthly household spending, household size, race, education, region, and baseline monthly spending of \$3,000. Standard errors in parentheses.

To quantify the extent to which individuals with faster habit decay rates adapt hedonically to the election outcome, we regress the measure of hedonic adaptation speed on a binary indicator for habit decay speed exceeding 50% per year. As reported in Column 1 of Table 5, respondents with habit decay speeds above this threshold exhibit, on average, a 1.31-point faster happiness adaptation rate, equivalent to a 40% increase relative to those with slower habit decay (below 50% per year). This finding remains robust when controlling for demographic characteristics and baseline spending levels from our survey, as shown in Column 2 of Table 5.

3.3 Welfare Impacts of Habit Formation and Peer Effects

This subsection details the measurement of two preference parameters expressed as ratios of utility derivatives, both of which are important for resolving the Easterlin paradox (Section 4.2). The first parameter, $-\frac{u_H}{u_C}$, is the slope of the indifference curve and governs the welfare impact of habit formation relative to one's own consumption. The second parameter, $\frac{u_{C_{\text{others}}}}{u_H}$, assesses the relative welfare impacts of peer effects and habit formation. As ratios of utility derivatives, they are naturally identified by trade-offs: habit against one's own consumption for $-\frac{u_H}{u_C}$, and peer effects against habit for $\frac{u_{C_{\text{others}}}}{u_H}$. To identify the preference parameters while preserving the generality of the utility function, we establish identification up to second-order approximations of the utility differences between choice alternatives, with approximation errors bounded by the remainders of the corresponding Taylor expansions.²² Exact identification, as

²²The approximation for $-\frac{u_H}{u_C}$ is of second order because the identifying variation is symmetric around the baseline, allowing second-order derivatives of the utility function to cancel out. Similarly, the approximation for $\frac{u_{C_{\text{others}}}}{u_H}$ is of first order, as the corresponding identifying variation lacks this symmetry. The alternative \$3,000 spending baseline, which induces



Figure 5: Spending Profiles for a Survey Question on Slope of Indifference Curve

established in Proposition 1, entails no approximation.²³

The slope of the indifference curve, $-\frac{u_H}{u_C}$, reflects the trade-off between habit and (future) spending. To identify it, the survey question uses a trade-off between past spending and future spending (Figure 5). Proposition 2 specifies how such variation identifies $-\frac{u_H}{u_C}$: preferring Universe One over Universe Two means that increasing future spending yields more utility gain than the utility loss from increasing past spending, and therefore u_C is more important than u_H (relative to the threshold $\frac{(\rho+\theta)\Delta C_{\text{future}}}{\rho\Delta C_{\text{past}}+\theta\Delta C_{\text{future}}}$), which explains the direction of the inequality in the proposition.

Proposition 2. *Under a second-order approximation, $-\frac{u_H}{u_C} < \frac{(\rho+\theta)\Delta C_{\text{future}}}{\rho\Delta C_{\text{past}}+\theta\Delta C_{\text{future}}}$ if a respondent chooses*

smaller spending differences, provides a useful gauge of the magnitude of the approximation errors. Reassuringly, Appendix D.1 shows that parameter estimates are highly robust to the choice of baseline.

²³The contrast in orders reflects what each parameter asks of the data: the decay rate is recovered from a comparison along a single margin—own habit at fixed future consumption—so monotonicity of u in H pins it down exactly, whereas the welfare parameters are trade-offs between two margins (habit against own consumption for $-u_H/u_C$, peer effects against habit for $u_{C_{\text{others}}}/u_H$) and are therefore identified only up to the order at which those margins are linearized.

*Universe One over Universe Two for a more satisfying future experience in an indifference-curve slope question.*²⁴

As the threshold depends on multiple parameters, the response distribution to this question alone is insufficient to fully characterize this parameter, necessitating the joint estimation of all relevant parameters. The estimated slope of the indifference curve is 0.63 (Table 2), implying that the habit intensity parameter for additive habit (everywhere) and multiplicative habit (in steady state) is about 0.6. This provides direct micro evidence for the parameter value often required to match macro data, a magnitude that observational micro-data estimates have frequently failed to yield (Havranek et al., 2017). This estimate indicates that, to a first-order approximation, around 60% of the utility derived from total consumption is eventually habituated. This suggests a substantial welfare impact, consistent with Van Praag and Frijters's (1999) finding that approximately 60% of income's effect on happiness is lost over time.

To assess the welfare consequence of peer effects relative to habit formation, we measure $\frac{u_{C_{\text{others}}}}{u_H}$. By accounting for the potential contemporaneous influence of others' spending (i.e., peer effects) in the utility function, $u(C, C_{\text{others}}, H)$, the ratio $\frac{u_{C_{\text{others}}}}{u_H}$ can be elicited by posing a trade-off between others' future spending and one's own past spending. Proposition 3 shows how such variation identifies $\frac{u_{C_{\text{others}}}}{u_H}$. Figure 6 presents the resulting survey question implementing the identifying variation. Preferring Universe One implies that peer effects matter relatively less than habit formation, explaining the direction of the inequality in the proposition.

Proposition 3. *Under a first-order approximation, $\frac{u_{C_{\text{others}}}}{u_H} < \frac{\rho}{\rho+\theta} \frac{\Delta C}{\Delta C_{\text{others}}}$ if a respondent chooses Universe One over Universe Two for a more satisfying future experience in a $\frac{u_{C_{\text{others}}}}{u_H}$ question.*²⁵

The point estimate for $\frac{u_{C_{\text{others}}}}{u_H}$ is 0.30, suggesting that habit formation has about three times the welfare impact of peer effects.²⁶ Combining this parameter with the slope of the indifference curve yields an estimate of $\frac{u_{C_{\text{others}}}}{u_C}$ of -0.19. This indicates that a one-dollar increase in others' spending is equivalent to a

²⁴In Figure 5, $\Delta C_{\text{past}} = \$2,000$ and $\Delta C_{\text{future}} = \200 . Table 4 details additional values of ΔC_{past} and ΔC_{future} .

²⁵In Figure 6, $\Delta C = \$2,000$ and $\Delta C_{\text{others}} = \200 . Additional values of ΔC and ΔC_{others} are detailed in Table 4.

²⁶This paper's model and estimation place no restriction on the sign of $u_{C_{\text{others}}}$, which may be positive (indicating, e.g., altruism) or negative (reflecting, e.g., peer comparison). The estimated sign of $u_{C_{\text{others}}}$ implies that the overall welfare impact of (total-consumption) peer effects is negative.

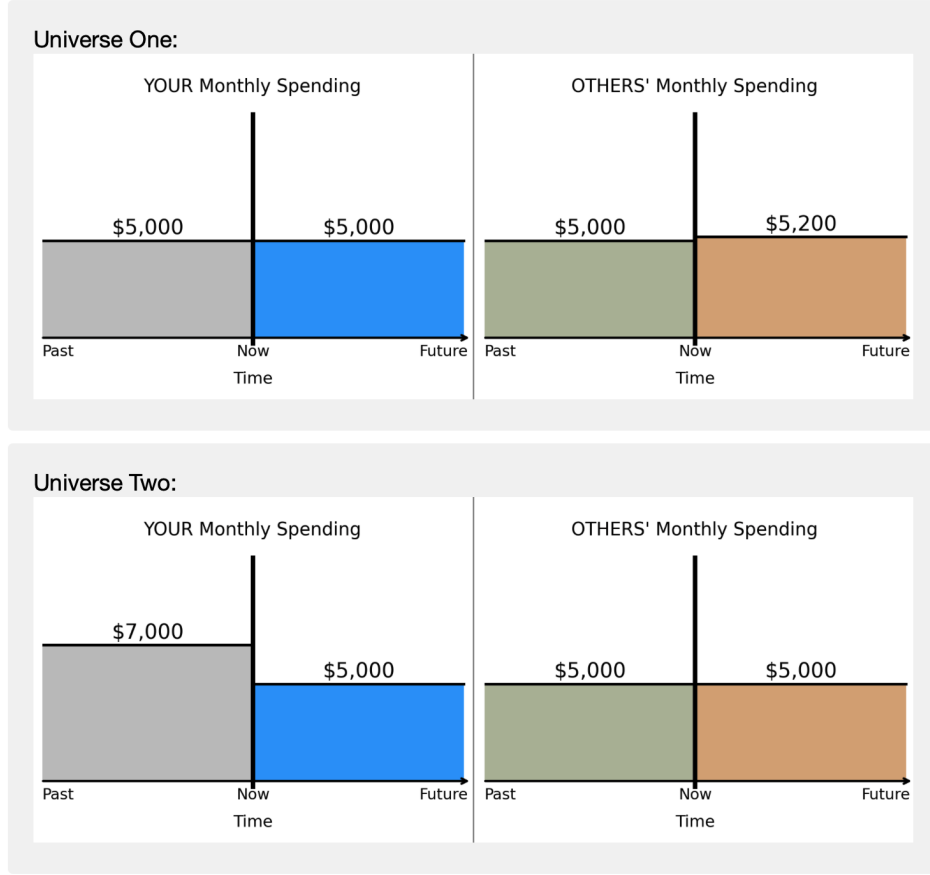


Figure 6: Spending Profiles for a Survey Question on $\frac{u_{C_{\text{others}}}}{u_H}$

20-cent decrease in one's own spending, which is consistent with existing estimates of the welfare impact of (consumption) peer effects (De Giorgi et al., 2020; Lewbel et al., 2022).

4 Applications

4.1 Impact of Habit Decay Speed

The speed of habit decay can significantly affect the performance of habit formation models. To illustrate, we evaluate the impact of habit decay speed in the widely cited habit formation framework of Campbell and Cochrane (1999), focusing on its implications for explaining the equity premium—a cornerstone application of habit formation.²⁷

²⁷ θ governs the linear habit evolution $\dot{H} = \theta(C - H)$, which is structurally identical whether the reference stock is built from one's own past consumption (internal habit) or others' (external habit). The two specifications differ only in the pricing kernel, as internal habit forces the agent to internalize consumption's effect on future habit. However, for power utility with a

Table 6: Impact of Habit Decay Speed

	Postwar	Habit formation		
	(1)	(2)	(3)	(4)
Habit decay factor	-	0.11	0.64	0.30
Expected excess ln return	6.69%	6.71%	41.11%	16.51%
Standard deviation of excess ln return	15.20%	15.64%	30.44%	22.01%
Sharpe ratio	0.43	0.43	1.35	0.75

Notes: Column 1: postwar (1947–95) value-weighted New York Stock Exchange stock-index returns. Column 2: Campbell and Cochrane (1999) calibration; 0.11 is the annual habit decay factor implied by their surplus-consumption-ratio persistence coefficient, ϕ . Column 3: this paper’s estimated habit decay factor. Column 4: a habit decay factor far below the 99% CI lower bound. Boldface indicates adjustments to Campbell and Cochrane’s (1999) calibration. All values annualized.

Theoretically, in simple additive habit formation models, faster habit decay reduces agents’ risk aversion because habit’s rapid adjustment to consumption alleviates concerns about maintaining habitual spending. Also employing an additive habit framework, Campbell and Cochrane (1999), however, have agents who are more risk averse when habit decays faster, as their (implied) steady-state habit intensity rises with the habit decay rate rather than remaining constant.²⁸ Higher habit intensity amplifies the risk that consumption fluctuations will fall below the habit-intensity-adjusted threshold, thereby elevating agents’ risk aversion. The overall effect of a higher habit decay rate in this framework reflects the interplay of these opposing dynamics, ultimately rendering agents more risk averse.

Quantitatively, Table 6 presents the relevant moments for the equity premium in observed data (Column 1) and in Campbell and Cochrane’s model under different habit decay speeds. Column 2 shows that their model matches the observed moments under their calibrated annual habit decay factor of 0.11. Column 3 shows that with the estimated habit decay factor of 0.64,²⁹ the model generates equity premiums that

constant interest rate and the linear habit evolution, this difference is exactly proportional across marginal utilities and therefore cancels out in asset pricing equations, which depend only on marginal-utility ratios. Campbell and Cochrane (1999) further confirm numerically that even under their nonlinear habit process, the average and standard deviation of excess return and Sharpe ratio are little changed by the internal-habit specification. As our application focuses strictly on these moments, applying our measured decay rate provides a valid quantitative illustration of θ ’s impact on the model’s asset-pricing predictions.

²⁸In this study’s data, responses to questions assessing the speed of habit decay exhibit negligible correlation with those gauging habit intensity. The average correlation coefficient is only -0.06.

²⁹This exercise holds Campbell and Cochrane’s remaining calibration fixed. Jointly re-tuning the other parameters might allow the model to match the equity premium under a faster habit decay speed, but the question pursued here is not whether Campbell and Cochrane can be re-fitted to data—it is the sensitivity of Campbell and Cochrane’s equity-premium prediction to this parameter. The sensitivity quantifies how much of the model’s asset-pricing performance was being identified by the slow habit decay rather than by the other parameters, and is informative for whether the calibrated decay speed should be interpreted as a preference parameter.

significantly exceed historical observations. This occurs because faster habit decay heightens agents' risk aversion, prompting them to demand unrealistically high returns to bear the historical level of consumption risk. Reducing the habit decay factor improves the alignment between the model moments and observed data, yet the percentage differences remain at least 40% even at a decay factor of 0.30 (Column 4), a speed well outside the 99% confidence interval for the habit decay factor.

The survey respondents may not fully represent the marginal investors who price assets. However, they need not mirror marginal investors in all characteristics. The preceding analysis remains robust provided the respondents' typical habit decay rate is close to that of marginal investors. This would hold if the habit decay rate is a fundamental preference parameter with minimal variation across demographic groups. Evidence presented in Appendix D.1 is consistent with this view, showing that the habit decay rate does not vary significantly with age, gender, household size, household income, historical or expected spending, race, education, or region.

Where, then, does the discrepancy between the survey-based decay rate and the much slower rate in Campbell and Cochrane's calibration come from? In that calibration, habit persistence is chosen to match the serial correlation of the log price-dividend ratio rather than directly measured as a preference parameter. The over-prediction in Columns 3 and 4 of Table 6 therefore suggests that the slow decay rate in Campbell and Cochrane's calibration reflects the targeted aggregate moments more than the underlying habit-formation preference.

The above analysis shows that the performance of a widely cited habit formation framework can be sensitive to the speed of habit decay. This result, however, warrants cautious interpretation. Given the evidence above that habit formation exists, the discrepancy suggests that existing models need refinement, not that modeling habit formation is misguided.

4.2 Explaining the Easterlin Paradox

The happiness–income paradox, proposed by Easterlin, states that income and happiness tend to be positively correlated in the short run and in the cross-section, but uncorrelated in the long run (Easterlin, 1973, 1974, 1995, 2001, 2017; Kaiser and Vendrik, 2019; Easterlin and O'Connor, 2025). Alternative views have been proposed, such as that U.S. data might be outliers (Stevenson and Wolfers, 2008), and that

life satisfaction can be time-intensive (Kimball and Willis, 2023). Despite the debate, the literature seems broadly in agreement that the empirical gradient of happiness with respect to income is small, and that the cross-section and short-run gradients tend to be larger than the long-run gradient, especially when the time horizon is long enough to cover full economic cycles (Easterlin and O'Connor, 2022). This section explores an explanation of the happiness–income pattern through the lens of habit formation and peer effects. To clarify the intuition behind this explanation, we consider a zero long-run gradient, noting that alternative perspectives can be accommodated with minor adjustments to parameter values without altering the intuition.

Habit formation and peer effects have been the most popular potential explanations for the paradox (Easterlin, 2001; Rayo and Becker, 2007; Clark et al., 2008). Recent evidence on peer effects (Luttmer, 2005; De Giorgi et al., 2020; Lewbel et al., 2022) suggests that they are not powerful enough to fully explain the phenomenon. To our knowledge, evidence on whether habit formation can help explain the paradox is absent from the literature. Based on the measurements above, this subsection argues that while habit formation alone is also not strong enough to generate the happiness–income pattern of the Easterlin paradox, combining it with peer effects can largely account for it.

Four clarifications merit discussion before proceeding. First, this section focuses on the causal mechanism by which income influences happiness. Everyday life experiences and studies exploiting exogenous variations support this perspective (Frijters et al., 2004; Gardner and Oswald, 2007). Factors affecting happiness other than income offer little help in explaining the paradox because they generally improve with income, making the long-run flat happiness–income gradient even more mysterious (Di Tella and MacCulloch, 2008). Second, this analysis aligns with the Benthamite tradition (Bentham, 1781; Perez-Truglia, 2020), positing that distinctions between happiness and utility exert negligible influence on their relationships with habit formation and peer effects. This assumption is grounded in the complex and ambiguous normative status of their differentiation in contemporary literature (Benjamin et al., 2024, 2014a,b, 2012; Kimball and Willis, 2023). Third, the paradox persists when substituting consumption for income because consumption is closely related to income (Carroll and Summers, 1991),³⁰ while happiness still has a long-term trend of about zero. Accordingly, the ensuing discussion regards income and consumption as interchange-

³⁰Real GDP per capita and real personal consumption expenditure in the United States have a correlation coefficient of 0.99 from 1947 to 2024.

able. Fourth, the literature identifies three measures of happiness: affect, capturing recent feelings; life satisfaction, reflecting an evaluation of life as a whole; and eudaimonia, encompassing personal growth and meaning. This study focuses on affect and life satisfaction due to their measurement reliability, extensive investigation, and direct relevance to the paradox. Consistent with this focus, [Layard and De Neve \(2023\)](#) argue that eudaimonia is better thought of as a contributor to, rather than a direct measure of, well-being. In the subsequent analysis, affect is proxied by instantaneous utility, and life satisfaction by lifetime utility.

To assess whether habit formation and peer effects can explain the Easterlin paradox, while ensuring robustness to potential misspecification in existing habit formation models (as suggested in the last subsection), we conduct semi-structural simulations grounded in this study’s evidence of both phenomena. In particular, the simulations specify agents as influenced by both internal habit formation and peer effects. Habit evolves according to equation (1), with the decay rate calibrated to its estimated value of 1.03. Peer effects arise after agents observe changes in others’ consumption, which is assumed to occur k years after those changes.³¹

Under first-order approximations, the effect of habit formation on utility is captured by the ratio $\frac{u_H}{u_C}$, while the impact of peer effects on utility is captured by $\frac{u_{C_{\text{others}}}}{u_C}$. As shown in Section 3.3, the estimates of $\frac{u_H}{u_C}$ and $\frac{u_{C_{\text{others}}}}{u_C}$ both exceed -1 at the 95% confidence level, indicating that habit formation and peer effects, considered separately, are insufficient to eliminate the long-run happiness gain from income growth.

The stylized zero long-run happiness–income gradient requires that

$$\frac{u_H}{u_C} + \frac{u_{C_{\text{others}}}}{u_C} = -1. \quad (2)$$

The point estimate of the left-hand side of the above equation is -0.82. While this is statistically distinguishable from the strict knife-edge of -1 at the 95% level, it implies that habit formation and peer effects combine to offset 82% of the initial happiness gains from permanent income growth, with a 95% confidence interval spanning a 69% to 96% offset. This leaves a residual gradient equal to just 18% of the short-run effect (95% CI [4%, 31%]). Rather than viewing this as a failure to hit a perfectly flat plateau, this 18% residual aligns with the perspective that habit formation and peer effects are the primary, but

³¹The exact value of k is irrelevant to the explanation, as it only affects the rate of convergence to the steady state.

not exclusive, drivers. It also leaves room for other proposed explanations—such as inequality, scale use, and economic growth’s externalities—to account for the remaining gap. To highlight the intuition behind an explanation based on habit formation and peer effects, the subsequent analysis focuses on the stylized benchmark in which these forces exactly offset the consumption gains (i.e., the left-hand side is -1).

The intuition of equation (2) is that, to a first-order approximation, habit formation and peer effects fully offset the hedonic impact of permanent changes in consumption over the long run. To see this, consider an economy initially in a steady state, in which residents maintain a constant level of happiness prior to time t_0 . Suppose that, from t_0 onward, the economy experiences growth, resulting in a permanent increase in each individual’s consumption by a small amount Δc (Figure 7a). Consequently, to the first-order approximation,³² at t_0 , residents’ happiness, as measured by affect, rises by $u_C \Delta c$. Over time, however, individuals gradually habituate to this elevated level of consumption, which diminishes affect (Figure 7b). At $t_0 + k$, individuals realize that others in the economy also enjoy the same elevated consumption level. This peer comparison further depresses affect and, in conjunction with habit formation, ultimately restores affect to its initial level. Similarly, life satisfaction exhibits an initial increase followed by a gradual decline, ultimately reverting to its prior steady-state level (Figure 7c). This dynamic, hereafter termed the *wear-off effect*, reflects the combined influence of habit formation and peer effects, which gradually offset the enhancement in happiness induced by a one-episode permanent growth in consumption.

In reality, economies exhibit sustained growth over time, resulting in progressively higher earnings and consumption for individuals. To capture this key aspect, suppose that each agent’s consumption rises permanently by Δc at the beginning of every year following t_0 , as depicted in Figure 7d. The corresponding evolution of the agent’s happiness is illustrated in Figures 7e and 7f. Within each year after t_0 , habit formation and peer effects diminish the happiness gains derived from consumption growth, consistent with the dynamics observed in the single-episode growth scenario. However, the long-term trajectory of happiness deviates significantly: rather than steadily reverting to its prior equilibrium, happiness incrementally rises and eventually plateaus at an elevated level. We label these two new patterns of happiness dynamics the *transition effect* and the *plateau effect*.

³²This section focuses on first-order approximations because the identification of $\frac{u_{C, \text{others}}}{u_C}$ is under a first-order approximation. This is a good approximation when Δc is small, which is maintained here.

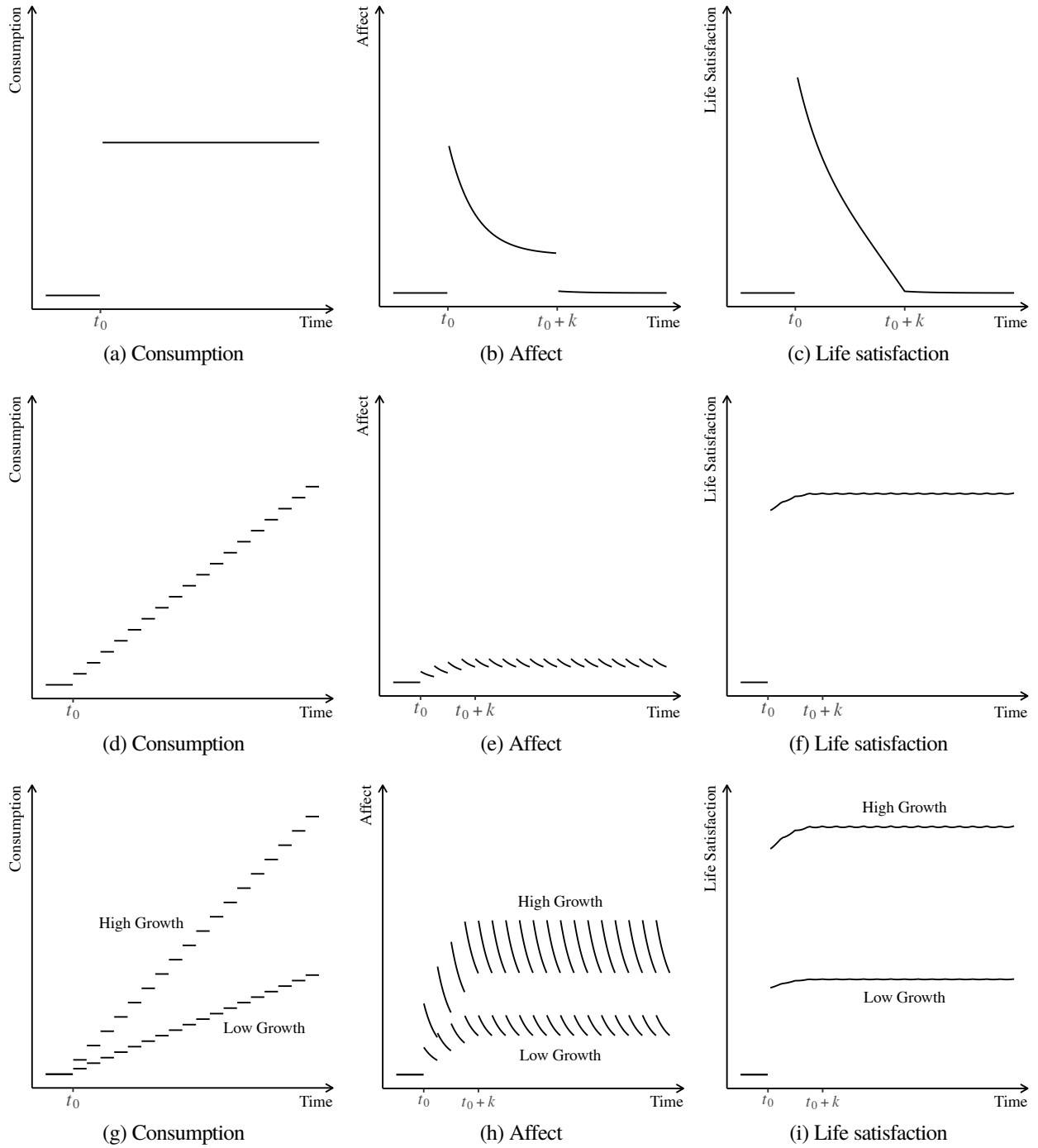


Figure 7: Happiness Simulations

Notes: Habit decay rate and time discount rate: 1.03 and 0.10, based on the estimates. For intuition, $\frac{u_H}{u_C} = -0.75$ and $\frac{u_{C_{\text{others}}}}{u_C} = -0.25$, preserving measured relative welfare importance and setting their sum to -1. Using exact point estimates (summing to -0.82) would give the plateaus a slight upward drift, reflecting the 18% residual gradient discussed above. To make the growth effect on affect visible, Panel (h)'s y-axis is set to 20% of Panel (i)'s. Discontinuities in Panels (e) and (h) arise because consumption is assumed to increase at the start of each year after t_0 ; continuous changes would smooth them without affecting the analysis.

The transition effect arises from the recurring nature of consumption growth. Each annual increment boosts happiness while initiating a new wear-off effect, both of which stack onto those from previous years. In the short run, this compounding process creates an upward trend in happiness, diverging from the downward trajectory typical of a one-episode growth. Over time, the countervailing forces of habit formation and peer effects progressively mitigate this happiness-enhancing momentum, eventually leading to the plateau effect, where happiness reaches a new steady state.

Because the wear-off effect is proportional to Δc , the transition and plateau effects are also proportional to Δc .³³ Consequently, more rapid consumption growth (Figure 7g) elicits stronger happiness responses (Figures 7h and 7i). This phenomenon, labeled the *growth effect*, implies that higher consumption growth induces more elevated happiness levels during both the transition and plateau phases.

The growth effect explains the positive cross-sectional correlation between income and happiness: higher income growth makes people or countries richer and places them on higher happiness trajectories. Economic fluctuations in reality cause consumption to fluctuate, frequently putting the agent into transition phases. The transition effect thus accounts for the short-term positive correlation between income and happiness. Whether income rises or falls, this effect always implies a positive association between income and happiness. The plateau effect explains the zero long-run correlation between income and happiness. Despite sustained income growth, happiness plateaus, yielding a zero correlation between happiness and income in the long run.³⁴

An alternative hypothesis for the Easterlin paradox posits that individuals' reporting scales change with income, rather than their actual happiness adapting to income (Deaton, 2008; Prati and Senik, 2024). To control for this rescaling channel, this study's survey employs a binary-choice format and was conducted over a brief period, minimizing the impact and likelihood of potential changes in respondents' scale-use tendencies. To more directly assess rescaling, the survey included visual calibration questions, as intro-

³³This is a direct implication of first-order approximations. Given positive marginal utility of consumption—a standard assumption in economics—the analysis holds for higher-order approximations: the utility difference between high and low consumption states, though likely reduced, remains positive, preserving qualitative implications.

³⁴This time-series plateau is distinct from the cross-sectional happiness–income gradient at high incomes studied by Killingsworth et al. (2023), who find no flattening for most respondents. The two are mutually consistent: in the habit mechanism here, the plateau emerges as habits adjust over time, so at any given moment higher-income individuals—whose habits have not yet caught up—can still report higher happiness.

duced by Benjamin et al. (2023), both before and after the core survey modules to measure scale-use consistency. Appendix Figure A.3 shows that respondents' scale-use shifters and stretchers before and after the core modules align closely along the 45-degree line, with an average correlation of 0.84. After adjusting for response errors based on the framework of Benjamin et al. (2023), the correlations indicate essentially perfect consistency, confirming that little rescaling occurred during the survey. This analysis suggests that habit formation and peer effects could explain the Easterlin paradox without rescaling.

How does the above explanation address the questions that motivated the paradox: Does money buy happiness (Easterlin, 1973), and does economic growth improve the human lot (Easterlin, 1974)? To phrase the questions more accurately: if people ultimately care only about happiness and happiness eventually stops increasing with economic growth, should societies continue promoting economic growth after happiness plateaus? The answer implied by the explanation is yes, because happiness would decrease if the economy were to grow at a slower rate. In other words, economic growth initially raises happiness and subsequently maintains it. Slower growth would reduce happiness and stabilize it at a lower level. Maintaining growth therefore remains important even after happiness gains diminish, as it prevents the erosion of well-being that would accompany an economic slowdown.

5 Conclusions

This study introduces a new measurement approach to provide micro-level evidence on habit formation in total-consumption behavior. We specify key parameters governing habit's existence, decay speed, and welfare implications, and formally derive their separate identification through total-consumption variations. By implementing the variations through survey experiments and validating the results in various ways, we provide evidence on habit formation that is robust to identification challenges in observational, laboratory, or field data. In particular, we find that habit forms internally, decays by two-thirds annually, and generates a welfare effect equivalent to 63% of that from consumption and three times that from peer effects.

Our evidence supports incorporating habit formation into economic models via the common preference-based approach, with habit formulated relative to one's own past consumption. Our application to the canonical model of Campbell and Cochrane (1999) shows that habit decay can materially affect the performance

of habit-based models, and our estimate suggests that this parameter should be disciplined around an annual rate of approximately two-thirds. The micro-level evidence also provides a habit intensity of approximately 0.6, aligning with values commonly used in macroeconomic and financial models. When peer effects are incorporated, the evidence favors calibrating their welfare impact to roughly one-third that of habit formation.

The measurements also imply that habit formation alone, or peer effects alone, cannot explain the East-erlin paradox, but their combination can account for most of it. People eventually habituate to their income and observe others' similar income changes, largely neutralizing the long-run happiness–income gradient. Before full habituation or realization of others' income changes, happiness moves in the same direction as income changes, explaining the short-run positive happiness–income gradient. Richer people and countries accumulate more happiness changes in the short run, accounting for the positive cross-sectional happiness–income gradient. This mechanism indicates that happiness may rise with increasing income, but only temporarily, before habit formation and peer effects stabilize happiness at a plateau. Although income may lack a long-run effect on happiness, sustained economic growth can remain important for maintaining the stabilized level of happiness. This underscores the important role of policies that promote sustained economic growth, even as the happiness gains from such growth vanish.

These results point to ways in which innovative measurement approaches can shed light on modern economic theories that are difficult to assess with conventional approaches, fostering the parallel development of new theories and new measurements.

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Online Appendices A–E for “Measuring Habit Formation”

The appendices present the proofs of identifying propositions, aggregation of preference parameters, response correlations, robustness checks, and additional survey questions.

A Proofs of Propositions

This appendix presents proofs of the propositions in the main text. It begins by deriving three quantities: the first (Ψ) is used in the proofs of Propositions 2 and 4, the second (Υ) in the proof of Proposition 4 (Appendix E.1), and the third (Φ) in the proof of Proposition 3. The proof of Proposition 1 uses only the habit-deviation paths $\Delta H(t)$ derived below, not the utility-difference expansions.

A.1 Three Quantities

This subsection derives three quantities regarding the utility differences between three classes of spending profiles used in the survey questions and the baseline steady state of $C(t) = H(t) = \bar{C} = \bar{H} \forall t$, facilitating the proofs of the identifying propositions.

To start, note that $\dot{H} = \theta(C - H)$ leads to

$$H(t) = e^{-\theta(t-t_0)} H(t_0) + \int_{t_0}^t e^{-\theta(t-s)} \theta C(s) ds,$$

which implies that the deviation of habit from its steady-state level is

$$\begin{aligned} \Delta H(t) &\equiv H(t) - \bar{H} \\ &= e^{-\theta(t-t_0)} (H(t_0) - \bar{H}) + \int_{t_0}^t e^{-\theta(t-s)} \theta (C(s) - \bar{H}) ds \\ &= e^{-\theta(t-t_0)} \Delta H(t_0) + \int_{t_0}^t e^{-\theta(t-s)} \theta \Delta C(s) ds, \end{aligned}$$

where $\Delta C(t) \equiv C(t) - \bar{C} = C(t) - \bar{H}$.

The first quantity. The first quantity pertains to the class of spending profiles characterized by $\Delta C(t) = A \cdot 1(t \leq 0) + B \cdot 1(t > 0)$. Under such spending profiles, for $t \geq 0$,

$$\begin{aligned}\Delta H(t) &= e^{-\theta\infty} \Delta H(-\infty) + \int_{-\infty}^0 e^{-\theta(t-s)} \theta A ds + \int_0^t e^{-\theta(t-s)} \theta B ds \\ &= e^{-\theta t} A + (1 - e^{-\theta t}) B.\end{aligned}\tag{3}$$

For the second equality, note that $\Delta H(-\infty)$ is finite by the definition of habit.

The difference between lifetime utilities starting now ($t = 0$), $\int_0^\infty e^{-\rho t} u(C(t), H(t)) dt$, under this class of spending profiles and under the steady state $(\bar{C}, \bar{H})$ is

$$\begin{aligned}\Psi(A, B) &\equiv \int_0^\infty e^{-\rho t} u(C(t), H(t)) dt - \int_0^\infty e^{-\rho t} u(\bar{C}, \bar{H}) dt \\ &= \int_0^\infty e^{-\rho t} [u(\bar{C} + B, \bar{H} + e^{-\theta t} A + (1 - e^{-\theta t}) B) - u(\bar{C}, \bar{H})] dt \\ &= \int_0^\infty e^{-\rho t} \left\{ u_C B + \frac{1}{2} u_{CC} B^2 + u_{CH} B (e^{-\theta t} A + (1 - e^{-\theta t}) B) + \dots \right. \\ &\quad \left. + \left[u_H (e^{-\theta t} A + (1 - e^{-\theta t}) B) + \frac{1}{2} u_{HH} (e^{-\theta t} A + (1 - e^{-\theta t}) B)^2 + \dots \right] \right\} dt,\end{aligned}$$

where in the last equality's expansion (and those below), “ $+\dots$ ” denotes the Taylor remainder beyond the displayed order. By Taylor's theorem and the assumed differentiability of u , this remainder is bounded uniformly in t —the spending deviations lie in a fixed compact neighborhood of the steady state—and integrates against $e^{-\rho t}$ without affecting the stated approximation order in Propositions 2–4.

The second quantity. The second quantity is related to the class of spending profiles characterized by

$$\Delta C(t) = A \cdot 1(-1 < t \leq 0) + B \cdot 1(0 < t \leq 1).$$

It follows that, for $t \geq 0$,

$$\begin{aligned}\Delta H(t) &= e^{-\theta\infty} \Delta H(-\infty) + \int_{-1}^0 e^{-\theta(t-s)} \theta A ds + \int_0^{\min\{1, t\}} e^{-\theta(t-s)} \theta B ds \\ &= e^{-\theta t} [(1 - e^{-\theta}) A + (e^{\theta \min\{1, t\}} - 1) B].\end{aligned}$$

The difference between lifetime utilities starting now, $\int_0^\infty e^{-\rho t} u(C(t), H(t)) dt$, under this class of spending profiles and under the steady state $(\bar{C}, \bar{H})$ is

$$\begin{aligned}
& \Upsilon(A, B) \\
& \equiv \int_0^\infty e^{-\rho t} u(C(t), H(t)) dt - \int_0^\infty e^{-\rho t} u(\bar{C}, \bar{H}) dt \\
& = \int_0^1 e^{-\rho t} [u(\bar{C} + B, \bar{H} + e^{-\theta t}((1 - e^{-\theta})A + (e^{\theta t} - 1)B)) - u(\bar{C}, \bar{H})] dt \\
& \quad + \int_1^\infty e^{-\rho t} [u(\bar{C}, \bar{H} + e^{-\theta t}(1 - e^{-\theta})(A + Be^\theta)) - u(\bar{C}, \bar{H})] dt \\
& = \int_0^1 e^{-\rho t} \left\{ u_C B + \frac{1}{2} u_{CC} B^2 + u_{CH} B e^{-\theta t} ((1 - e^{-\theta})A + (e^{\theta t} - 1)B) + \dots \right. \\
& \quad \left. + \left[u_H e^{-\theta t} ((1 - e^{-\theta})A + (e^{\theta t} - 1)B) + \frac{1}{2} u_{HH} (e^{-\theta t} ((1 - e^{-\theta})A + (e^{\theta t} - 1)B))^2 + \dots \right] \right\} dt \\
& \quad + \int_1^\infty e^{-\rho t} \left[u_H e^{-\theta t} (1 - e^{-\theta})(A + Be^\theta) + \frac{1}{2} u_{HH} (e^{-\theta t} (1 - e^{-\theta})(A + Be^\theta))^2 + \dots \right] dt.
\end{aligned}$$

The third quantity. When others' spending varies, it is necessary to consider peer effects and the class of spending profiles with $\Delta C(t) = A \cdot 1(t \leq 0) + B \cdot 1(t > 0)$ and $\Delta C_{\text{others}}(t) = D \cdot 1(t \leq 0) + E \cdot 1(t > 0)$.

The difference between lifetime utilities starting now, $\int_0^\infty e^{-\rho t} u(C(t), C_{\text{others}}(t), H(t)) dt$, under this class of spending profiles and under the steady state $(\bar{C}, \bar{C}_{\text{others}}, \bar{H})$ —the third quantity—is

$$\begin{aligned}
& \Phi(A, B, D, E) \\
& \equiv \int_0^\infty e^{-\rho t} u(C(t), C_{\text{others}}(t), H(t)) dt - \int_0^\infty e^{-\rho t} u(\bar{C}, \bar{C}_{\text{others}}, \bar{H}) dt \\
& = \int_0^\infty e^{-\rho t} [u(\bar{C} + B, \bar{C}_{\text{others}} + E, \bar{H} + e^{-\theta t}A + (1 - e^{-\theta t})B) - u(\bar{C}, \bar{C}_{\text{others}}, \bar{H})] dt \\
& = \int_0^\infty e^{-\rho t} \{ u_C B + u_{C_{\text{others}}} E + u_H [e^{-\theta t}A + (1 - e^{-\theta t})B] \\
& \quad + \frac{1}{2} u_{CC} B^2 + \frac{1}{2} u_{C_{\text{others}} C_{\text{others}}} E^2 + \frac{1}{2} u_{HH} [e^{-\theta t}A + (1 - e^{-\theta t})B]^2 \\
& \quad + u_{CC_{\text{others}}} BE + u_{CH} B [e^{-\theta t}A + (1 - e^{-\theta t})B] \\
& \quad + u_{C_{\text{others}} H} E [e^{-\theta t}A + (1 - e^{-\theta t})B] + \dots \} dt.
\end{aligned}$$

A.2 Proof of Proposition 1

Proposition 1. $\theta > -\ln \left(1 - \frac{\Delta C_{U1}}{\Delta C_{U2}}\right)$ if a respondent chooses Universe One over Universe Two for a more satisfying future experience in a habit decay rate question.

Proof. Conditional on the existence of internal habit formation (Section 3.1), $u_H < 0$ along the relevant range of H . In a habit decay rate question, future spending equals the baseline in both universes, so the two universes differ only in the habit inherited from past spending. Universe One sustains a past deviation ΔC_{U1} and Universe Two concentrates a past deviation ΔC_{U2} in the most recent year. By the habit-deviation expressions derived in Appendix A.1, the resulting future habit paths are, for $t \geq 0$,

$$H_{U1}(t) = \bar{H} + e^{-\theta t} \Delta C_{U1}, \quad H_{U2}(t) = \bar{H} + e^{-\theta t} (1 - e^{-\theta}) \Delta C_{U2},$$

so $H_{U1}(t) - H_{U2}(t) = e^{-\theta t} [\Delta C_{U1} - (1 - e^{-\theta}) \Delta C_{U2}]$ has constant sign in t .

A respondent preferring Universe One for a better future experience implies

$$\begin{aligned} 0 &< U(\text{Universe One}) - U(\text{Universe Two}) \\ &= \int_0^\infty e^{-\rho t} [u(\bar{C}, H_{U1}(t)) - u(\bar{C}, H_{U2}(t))] dt \\ &= \int_0^\infty e^{-\rho t} u_H(\bar{C}, \xi_t) e^{-\theta t} [\Delta C_{U1} - (1 - e^{-\theta}) \Delta C_{U2}] dt, \end{aligned}$$

where the last equality applies the mean value theorem with ξ_t between $H_{U1}(t)$ and $H_{U2}(t)$, and the integral converges because the integrand is bounded and discounted at rate $\rho > 0$. Since $u_H(\bar{C}, \xi_t) < 0$ and $e^{-\rho t}, e^{-\theta t} > 0$, the integrand has the sign of $-\Delta C_{U1} + (1 - e^{-\theta}) \Delta C_{U2}$ for every t . The difference is therefore positive only if

$$\Delta C_{U1} - (1 - e^{-\theta}) \Delta C_{U2} < 0,$$

which, since $0 < \Delta C_{U1} < \Delta C_{U2}$ (Table 4), is equivalent to

$$\theta > -\ln \left(1 - \frac{\Delta C_{U1}}{\Delta C_{U2}}\right).$$

□

A.3 Proof of Proposition 2

Proposition 2. *Under a second-order approximation, $-\frac{u_H}{u_C} < \frac{(\rho+\theta)\Delta C_{\text{future}}}{\rho\Delta C_{\text{past}}+\theta\Delta C_{\text{future}}}$ if a respondent chooses Universe One over Universe Two for a more satisfying future experience in an indifference-curve slope question.*

Proof. A respondent preferring Universe One for a better future experience in an indifference-curve slope question implies

$$\begin{aligned}
& 0 < U(\text{Universe One}) - U(\text{Universe Two}) \\
&= [U(\text{Universe One}) - U(\text{Baseline})] - [U(\text{Universe Two}) - U(\text{Baseline})] \\
&= \Psi(\Delta C_{\text{past}}, \Delta C_{\text{future}}) - \Psi(-\Delta C_{\text{past}}, -\Delta C_{\text{future}}) \\
&= \frac{1}{\rho} \left\{ u_C \Delta C_{\text{future}} + u_H \frac{\rho \Delta C_{\text{past}} + \theta \Delta C_{\text{future}}}{\rho + \theta} + \frac{1}{2} [u_{CC} (\Delta C_{\text{future}})^2 \right. \\
&\quad + 2u_{CH} \frac{\rho \Delta C_{\text{past}} \Delta C_{\text{future}} + \theta (\Delta C_{\text{future}})^2}{\rho + \theta} \\
&\quad \left. + u_{HH} \frac{\rho(\rho + \theta) (\Delta C_{\text{past}})^2 + 2\rho\theta \Delta C_{\text{past}} \Delta C_{\text{future}} + 2\theta^2 (\Delta C_{\text{future}})^2}{(\rho + \theta)(\rho + 2\theta)} \right\} \\
&\quad - \frac{1}{\rho} \left\{ u_C (-\Delta C_{\text{future}}) + u_H \frac{\rho(-\Delta C_{\text{past}}) + \theta(-\Delta C_{\text{future}})}{\rho + \theta} \right. \\
&\quad + \frac{1}{2} \left[u_{CC} (\Delta C_{\text{future}})^2 + 2u_{CH} \frac{\rho \Delta C_{\text{past}} \Delta C_{\text{future}} + \theta (\Delta C_{\text{future}})^2}{\rho + \theta} \right. \\
&\quad \left. \left. + u_{HH} \frac{\rho(\rho + \theta) (\Delta C_{\text{past}})^2 + 2\rho\theta \Delta C_{\text{past}} \Delta C_{\text{future}} + 2\theta^2 (\Delta C_{\text{future}})^2}{(\rho + \theta)(\rho + 2\theta)} \right] \right\} \\
&= \frac{2}{\rho} \left(u_C \Delta C_{\text{future}} + u_H \frac{\rho \Delta C_{\text{past}} + \theta \Delta C_{\text{future}}}{\rho + \theta} \right),
\end{aligned}$$

where the third equality holds under a second-order approximation.

The inequality, by $u_C > 0$ and $\rho > 0$,¹ implies

$$-\frac{u_H}{u_C} < \frac{(\rho + \theta) \Delta C_{\text{future}}}{\rho \Delta C_{\text{past}} + \theta \Delta C_{\text{future}}}.$$

□

¹ $\Delta C_{\text{past}} > 0$ and $\Delta C_{\text{future}} > 0$ in the survey (Table 4).

A.4 Proof of Proposition 3

Proposition 3. *Under a first-order approximation, $\frac{u_{C_{\text{others}}}}{u_H} < \frac{\rho}{\rho+\theta} \frac{\Delta C}{\Delta C_{\text{others}}}$ if a respondent chooses Universe One over Universe Two for a more satisfying future experience in a $\frac{u_{C_{\text{others}}}}{u_H}$ question.*

Proof. A respondent preferring Universe One for a better future experience in a $\frac{u_{C_{\text{others}}}}{u_H}$ question implies

$$\begin{aligned}
0 &< U(\text{Universe One}) - U(\text{Universe Two}) \\
&= [U(\text{Universe One}) - U(\text{Baseline})] - [U(\text{Universe Two}) - U(\text{Baseline})] \\
&= \Phi(0, 0, 0, \Delta C_{\text{others}}) - \Phi(\Delta C, 0, 0, 0) \\
&= \int_0^\infty e^{-\rho t} u_{C_{\text{others}}} \Delta C_{\text{others}} dt - \int_0^\infty e^{-(\rho+\theta)t} u_H \Delta C dt \\
&= \frac{1}{\rho} u_{C_{\text{others}}} \Delta C_{\text{others}} - u_H \frac{1}{\rho+\theta} \Delta C,
\end{aligned}$$

where the third equality holds under a first-order approximation.

The last inequality, by $u_H < 0$ and $\rho > 0$,² implies

$$\frac{u_{C_{\text{others}}}}{u_H} < \frac{\rho}{\rho+\theta} \frac{\Delta C}{\Delta C_{\text{others}}}.$$

□

B Aggregation

What do the preference parameters of individual respondents tell us about the preference parameters of a representative agent? This question is of particular interest because almost all current models with habit formation assume a representative agent. This appendix shows that, given its existence, the representative agent's preference parameters are averages of the individuals' preference parameters.

To aggregate individuals, their welfare needs to be mutually comparable (comparability), and the representative agent's welfare should represent the average of the individuals' welfare (representativeness). To formalize the idea of comparability, we assume that at the homogeneous steady state of $\bar{C}_i = \bar{H}_i = \varphi \forall i$,

²The sign of u_H is elicited in the existence of internal habit formation (Section 3.1). $\Delta C_{\text{others}} > 0$ in the survey (Table 4).

spending an extra dollar while holding habit constant brings the same marginal utility to every individual:
 $u_{i,C}(\bar{C}_i, \bar{H}_i) = u_{j,C}(\bar{C}_j, \bar{H}_j) = \bar{u} \forall i, j.$

With the comparability of individuals' utilities, the representativeness of the representative agent (R) implies that $Nu_R(C_R, H_R) = \sum_i u_i(C_i, H_i)$ when $C_R = C_i = A$ and $H_R = H_i = B \forall i$ and $\forall A, B$ in the domains of the utility functions, where N is the number of individuals in the economy. That is, when the heterogeneities in behaviors (i.e., consumption and habit) are homogenized, the representative agent is the average individual agent in terms of welfare. To see what this condition means, first note that the difference between a representative-agent model and a heterogeneous-agent model is that in the former, everyone in the economy is the same, while in the latter, each individual can be different. If everyone in the heterogeneous economy becomes the same (i.e., homogeneous in consumption, habit, utility function, etc.), then the representative-agent model should behave the same as the homogenized heterogeneous-agent model, which results in the equality of $Nu_R(C_R, H_R) = \sum_i u_i(C_i, H_i)$ under $C_R = C_i, H_R = H_i$, and $u_R = u_i \forall i$. Now, allowing individuals to be heterogeneous along the dimension of the utility function after the normalization of the comparability condition, the representativeness condition simply formalizes the requirement that the representative agent represents the individuals along the welfare dimension, after controlling for consumption and habit.

B.1 Aggregation of Habit Decay Rate

Even though the habit decay rate (θ) and habit (H) are mapped one-to-one at each instant prior to reaching a steady state for any given consumption profile, there exist infinitely many pairs of θ and H that can simultaneously satisfy the representative agent's habit evolution equation ($\dot{H}_R = \theta_R(C_R - H_R)$), the individuals' habit evolution equation ($\dot{H}_i = \theta_i(C_i - H_i)$), and the comparability condition. The intuition is that while habit depends on the habit decay rate, its steady-state level does not. In other words, a different value of θ leads to a different value of H at every instant before the steady state, but this difference vanishes once the steady state is reached. The representativeness condition eliminates this indeterminacy and determines a unique θ_R given the individuals' θ_i 's.

To find the mapping between the aggregate habit decay rate (θ_R) and the individuals' habit decay rates (θ_i 's), imagine that everyone starts at the homogeneous steady state and increases their consumption by the

same small amount. That is, starting from $C_i = C_j = C_R = H_i = H_j = H_R \forall i, j$, increase consumption by the infinitesimal amount $\Delta C_i = \Delta C_j = \Delta C_R \forall i, j$. The resulting changes in utilities are

$$\begin{aligned}\Delta u_R(C_R, H_R) &= u_{R,C}(C_R, H_R) \Delta C_R + u_{R,H}(C_R, H_R) \Delta H_R \\ &= u_{R,C}(C_R, H_R) \Delta C_R + u_{R,H}(C_R, H_R) \theta_R \Delta C_R\end{aligned}$$

and

$$\begin{aligned}\Delta u_i(C_i, H_i) &= u_{i,C}(C_i, H_i) \Delta C_i + u_{i,H}(C_i, H_i) \Delta H_i \\ &= u_{i,C}(C_i, H_i) \Delta C_i + u_{i,H}(C_i, H_i) \theta_i \Delta C_i,\end{aligned}$$

where $u_{X,Y} \equiv \partial u_X / \partial Y$.

Since $N \Delta u_R(C_R, H_R) = \sum_i \Delta u_i(C_i, H_i)$, as implied by the representativeness condition,

$$u_{R,C}(C_R, H_R) \Delta C_R + u_{R,H}(C_R, H_R) \theta_R \Delta C_R = \frac{1}{N} \sum_i [u_{i,C}(C_i, H_i) \Delta C_i + u_{i,H}(C_i, H_i) \theta_i \Delta C_i].$$

Because $\Delta C_R = \Delta C_i$ and $u_{R,C}(C_R, H_R) = u_{i,C}(C_i, H_i) \forall i$ (see Appendix B.2),

$$\frac{u_{R,H}(C_R, H_R)}{u_{R,C}(C_R, H_R)} \theta_R = \frac{1}{N} \sum_i \frac{u_{i,H}(C_i, H_i)}{u_{i,C}(C_i, H_i)} \theta_i = \frac{1}{N} \sum_i \frac{u_{i,H}(C_i, H_i)}{u_{i,C}(C_i, H_i)} \cdot \frac{1}{N} \sum_i \theta_i,$$

where the second equality holds because of the independence between the slope of the indifference curve and the habit decay rate. With $u_{R,H}(C_R, H_R) = \frac{1}{N} \sum_i u_{i,H}(C_i, H_i)$ (see Appendix B.2) and $u_{R,C}(C_R, H_R) = u_{i,C}(C_i, H_i) \forall i$, it follows that

$$\theta_R = \frac{1}{N} \sum_i \theta_i.$$

That is, the representative agent's habit decay rate equals the average of the individual agents' habit decay rates.

B.2 Aggregation of Ratios of Utility Derivatives

In this appendix, we derive the relationships between the utility derivatives of the representative agent and those of the heterogeneous agents at the homogeneous steady state ($\bar{C}_R = \bar{H}_R = \bar{C}_i = \bar{H}_i \forall i$).

Because $Nu_R(C_R, H_R) = \sum_i u_i(C_i, H_i)$ when $C_R = C_i = A$ and $H_R = H_i = B \forall i$ and $\forall A, B$ in the domains of the utility functions, the utility derivatives of the representative agent are the averages of the utility derivatives of the individuals:

$$u_{R,X}(C_R, H_R) = \frac{1}{N} \sum_i u_{i,X}(C_i, H_i),$$

where X denotes the variable of differentiation (e.g., C, H).

Next, we derive the relationships between the ratios of the utility derivatives of the representative agent and those of the heterogeneous agents at the steady state, in three steps.

1. Under the normalization of $u_{R,C}(\bar{C}_R, \bar{H}_R) = u_{i,C}(\bar{C}_i, \bar{H}_i) = \bar{u} \forall i$,

$$\frac{1}{N} \sum_i u_{i,H} = -\frac{1}{N} \sum_i \left(-\frac{u_{i,H}}{u_{i,C}} \cdot u_{i,C} \right) = - \left[\frac{1}{N} \sum_i \left(-\frac{u_{i,H}}{u_{i,C}} \right) \right] \cdot u_{i,C} \equiv -\mu_{-\frac{u_{i,H}}{u_{i,C}}} \cdot \bar{u},$$

where $\mu_{-\frac{u_{i,H}}{u_{i,C}}} \equiv \frac{1}{N} \sum_i \left(-\frac{u_{i,H}}{u_{i,C}} \right)$. Similar notation is used hereafter.

2. Because the parameters are independent,

$$\frac{1}{N} \sum_i u_{i,C_{\text{others}}} = \frac{1}{N} \sum_i \left(\frac{u_{i,C_{\text{others}}}}{u_{i,H}} \cdot u_{i,H} \right) = \frac{1}{N} \sum_i \frac{u_{i,C_{\text{others}}}}{u_{i,H}} \cdot \frac{1}{N} \sum_i u_{i,H}.$$

3. With the above, the parameters of the representative agent can be calculated as follows:

$$-\frac{u_{R,H}}{u_{R,C}} = -\frac{\frac{1}{N} \sum_i u_{i,H}}{\frac{1}{N} \sum_i u_{i,C}} = -\frac{\frac{1}{N} \sum_i u_{i,H}}{\bar{u}} = \mu_{-\frac{u_{i,H}}{u_{i,C}}},$$

and

$$\frac{u_{R,C_{\text{others}}}}{u_{R,H}} = \frac{\frac{1}{N} \sum_i u_{i,C_{\text{others}}}}{\frac{1}{N} \sum_i u_{i,H}} = \frac{\frac{1}{N} \sum_i \frac{u_{i,C_{\text{others}}}}{u_{i,H}} \cdot \frac{1}{N} \sum_i u_{i,H}}{\frac{1}{N} \sum_i u_{i,H}} = \mu_{\frac{u_{i,C_{\text{others}}}}{u_{i,H}}}.$$

In summary, the representative agent's ratios of utility derivatives equal the averages of the individuals' ratios of utility derivatives.

C Response Correlations

Table A.1 presents the correlation matrix of survey responses across questions and waves. The results reveal generally low correlations among survey questions, indicating that most parameters of interest

Table A.1: Correlation Matrix of Survey Responses

Question	Wave	Internal		External		θ		$-\frac{u_H}{u_C}$		$\frac{u_{C, \text{others}}}{u_H}$	
		1	2	1	2	1	2	1	2	1	2
Internal	1	1.00	0.56	0.06	0.01	0.02	0.09	-0.49	-0.34	0.35	0.21
	2	0.56	1.00	0.07	0.06	0.08	0.05	-0.37	-0.54	0.32	0.31
External	1	0.06	0.07	1.00	0.18	0.02	0.01	-0.05	-0.03	-0.01	0.08
	2	0.01	0.06	0.18	1.00	0.00	-0.03	0.00	-0.03	0.01	0.01
θ	1	0.02	0.08	0.02	0.00	1.00	0.38	-0.01	-0.07	-0.02	0.00
	2	0.09	0.05	0.01	-0.03	0.38	1.00	-0.06	-0.11	0.01	0.05
$-\frac{u_H}{u_C}$	1	-0.49	-0.37	-0.05	0.00	-0.01	-0.06	1.00	0.45	-0.33	-0.20
	2	-0.34	-0.54	-0.03	-0.03	-0.07	-0.11	0.45	1.00	-0.28	-0.31
$\frac{u_{C, \text{others}}}{u_H}$	1	0.35	0.32	-0.01	0.01	-0.02	0.01	-0.33	-0.28	1.00	0.39
	2	0.21	0.31	0.08	0.01	0.00	0.05	-0.20	-0.31	0.39	1.00

Notes: Correlations of survey responses across questions and waves; baselines pooled. Internal and External: binary indicators equal to 1 if corresponding habit exists and 0 otherwise. Other responses range from 1 to 4, corresponding to U1U1, U1U2, U2U1, and U2U2. For θ , higher response values indicate lower parameter values. For $-\frac{u_H}{u_C}$ and $\frac{u_{C, \text{others}}}{u_H}$, higher response values indicate higher parameter values.

capture distinct facets of preference. However, several noteworthy exceptions emerge. The internal-habit response is negatively correlated with the response for $-\frac{u_H}{u_C}$ and positively correlated with the response for $\frac{u_{C, \text{others}}}{u_H}$, while the two ratio responses are negatively correlated with each other. Because these are correlations of discrete survey responses rather than correlations of individual-level structural parameters, they should be interpreted descriptively. In particular, the correlations combine heterogeneity in preferences with response error and need not coincide with the deterministic mapping implied by internally consistent choices under the maintained model. They nevertheless motivate a robustness check that relaxes the benchmark independence assumption: allowing the four continuous parameters to have an unrestricted within-respondent correlation matrix leads to estimates that remain close to the benchmark estimates, especially $\frac{u_{C, \text{others}}}{u_C} = -0.24$ (95% CI: [-0.34,-0.14]), compared with -0.19 ([-0.30,-0.08]) in the benchmark; and $\frac{u_H}{u_C} + \frac{u_{C, \text{others}}}{u_C} = -0.83$ (95% CI: [-0.95,-0.72]), compared with -0.82 ([-0.96,-0.69]) in the benchmark. These differences are not statistically distinguishable at the 5% level.

The test-retest correlations indicate moderate temporal stability, averaging 0.45 across all main questions. This excludes the question on the existence of external habit formation, as the absence of external habit formation implies random responses and a naturally low correlation. Ranging from 0.38 to 0.56, these test-retest correlations are consistent with those of typical choice-based experimental measures

of economic preferences (Chuang and Schechter, 2015; Meier and Sprenger, 2015). The variability in these correlations across questions may be due to response errors, the complexity or novelty of individual questions, or underlying heterogeneity in preferences.

D Robustness

This appendix evaluates the robustness of the findings across several dimensions, including demographics, spending baselines, time horizons, response biases and errors with nonzero and wave-varying means, first-wave-only estimation, strict ceteris-paribus compliance, and additional attention checks.

D.1 Effects of Demographics and Spending Baselines

The survey gathers demographic data on respondents' age, gender, household income and spending, household size, race, education, and region. Table A.2 presents the sample's demographic distributions alongside those of the 2020 U.S. Census. Incorporating these demographic variables as shifters of the parameter distributions in the statistical model reveals that none of their effects are statistically significant at conventional confidence levels after adjusting for multiple hypothesis testing (Table A.3). This is consistent with the literature's finding that demographics generally explain very little of the variation in deep structural parameters measured through experiments (Gillen et al., 2019; Falk et al., 2018).

Some respondents were randomized into a survey version with a baseline spending level of \$3,000, aligning more closely with their real-life spending. To assess the impact of baseline spending, we included an indicator for the \$3,000 baseline in the regressions. The results suggest that the baseline spending does not significantly impact θ and $\frac{u_{C_{\text{others}}}}{u_H}$, and exerts a marginally significant impact on $-\frac{u_H}{u_C}$ after false discovery adjustment (Table A.3). Aside from statistical precision considerations, the negative impact of the baseline on $-\frac{u_H}{u_C}$ potentially reflects the parameter's direct dependence on C and is consistent with diminishing marginal utility of consumption. It is worth noting that the respondents' real-life historical and expected spending do not significantly affect $-\frac{u_H}{u_C}$, suggesting that our respondents effectively engaged with the hypothetical scenarios in the experiments, providing responses unconfounded by their real-life circumstances. This highlights the effectiveness of our approach in inducing desired variations.

To further investigate how deviations of baseline spending from real-life spending might affect our

Table A.2: Respondent Demographics

Demographic	Category	First wave	Second wave	Census
Gender	Male	48.7%	47.3%	49.0%
	Female	51.3	52.7	51.0
Age	18–29	21.0	17.6	20.6
	30–39	20.1	20.9	17.5
	40–49	17.0	19.3	16.0
	50–64	30.0	31.0	24.9
	65 and older	11.9	11.2	21.1
Highest education level completed	High school graduate or less	9.9	9.5	37.9
	Some college	32.3	31.8	27.1
	Bachelor’s degree	38.0	37.5	22.2
	Graduate degree	19.8	21.2	12.8
Household income	Less than \$30,000	16.8	18.1	22.1
	\$30,000–\$49,999	15.1	16.2	15.7
	\$50,000–\$69,999	17.4	17.1	13.4
	\$70,000–\$89,999	12.7	12.9	10.8
	\$90,000–\$119,999	25.3	24.8	12.0
	\$120,000 and above	12.7	11.0	26.0
Race	White	68.4	69.9	61.6
	Black	11.1	10.1	12.1
	Hispanic or Latino	8.6	9.3	14.9
	Asian	6.7	5.5	5.9
	Other	5.2	5.2	5.5
Household size	1	20.4	21.8	28.5
	2	33.3	33.3	35.0
	3	20.0	19.3	15.0
	4 and above	26.3	25.6	21.5
Region	Midwest	18.7	18.6	20.8
	Northeast	17.0	17.0	17.4
	South	41.1	39.3	38.1
	West	22.6	24.8	23.7
	Outside U.S.	0.5	0.3	–
Observations		1,367	730	

Notes: Sources: author’s surveys and 2020 U.S. Census. Entries are percentages. “Outside U.S.”: respondents whose geolocated country is not the United States; no comparable Census category is available (“–”).

estimates, we allow the parameter estimates to differ for each baseline in our statistical model and estimate the modified model using our sample of respondents with real-life monthly household spending less than \$4,000. Each of these respondents was randomized with a 50-50 chance into either baseline. Column 1 of Table A.4 shows that none of the parameter estimates are statistically significantly different between the baselines at the 99% level.

Table A.3: Demographic Effects on Parameter Estimates

Demographic	θ	$-u_H/u_C$	$u_{C_{\text{others}}}/u_H$
Demeaned age	0.00 (0.00)	0.00 (0.00)	0.02 (0.01)
Demeaned age ²	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Male	-0.27 (0.14)	0.00 (0.07)	0.17 (0.19)
Log(household income)	0.00 (0.12)	-0.06 (0.06)	0.03 (0.17)
Log(past household monthly spending)	0.41 (0.22)	0.06 (0.11)	0.34 (0.29)
Log(expected household monthly spending)	-0.42 (0.22)	-0.09 (0.11)	-0.24 (0.28)
Household size	0.11 (0.05)	-0.01 (0.03)	0.01 (0.07)
Black	0.33 (0.23)	0.01 (0.12)	0.73 (0.30)
Hispanic	0.34 (0.26)	-0.13 (0.14)	0.56 (0.33)
Asian	-0.14 (0.29)	-0.15 (0.15)	0.63 (0.38)
Other race	-0.06 (0.32)	-0.06 (0.16)	-0.18 (0.42)
College graduate	-0.14 (0.15)	0.14 (0.08)	-0.16 (0.20)
West	0.31 (0.21)	0.19 (0.11)	-0.05 (0.29)
South	0.12 (0.19)	0.16 (0.10)	0.15 (0.26)
Northeast	0.00 (0.23)	0.03 (0.12)	-0.12 (0.31)
Outside of U.S.	0.59 (1.02)	0.12 (0.52)	0.29 (1.31)
Baseline=\$3,000	0.18 (0.16)	-0.23 [†] (0.09)	0.13 (0.21)
Observations	2,097	2,097	2,097

Notes: Standard errors in parentheses. [†], ^{††}, and ^{†††}: false-discovery-rate significance at the 90%, 95%, and 99% levels, using the Benjamini–Hochberg procedure.

Assessing the same issue from a different angle, we restrict our sample to respondents with real-life monthly spending within \$1,000 of the baselines and re-estimate the parameters of interest. The resulting estimates, shown in Column 2 of Table A.4, are essentially the same as our benchmark estimates in Table 2. Overall, the above evidence suggests that our parameter estimates are robust to deviations between our baseline spending and respondents’ real-life spending.

D.2 Finite Horizon

The model outlined in Section 2.1 adopts an infinite horizon, consistent with nearly all contemporary habit formation models in the literature. To assess the impact of this assumption, we rederive the identifying propositions for the preference parameters under finite horizons and find that the resulting changes are negligible. Specifically, the thresholds for the existence of internal habit formation, external habit formation, and the habit decay rate remain identical across both infinite and finite horizons. For other parameters, the thresholds adjust by replacing 1 with $1 - e^{-\rho T}$, $1 - e^{-(\rho+\theta)T}$, or $1 - e^{-(\rho+2\theta)T}$, which closely approximate

Table A.4: Robustness Check Estimates

Parameter	Difference between baselines (1)	Actual spending close to baseline spending (2)	Finite horizon (3)	Flexible response bias and error (4)	First-wave data (5)	Strict ceteris-paribus compliance (6)	Instruction manipulation checks (7)	Identical past checks (8)	No polar response (9)
$\text{sgn}(Q_{HI})$	0	-1	-1	-1	-1	-1	-1	-1	-1
	{0}	{-1}	{-1}	{-1}	{-1}	{-1}	{-1}	{-1}	{-1}
$\text{sgn}(Q_{HE})$	0	1	1	1	1	1	1	1	1
θ	{-2, -1, 0, 1, 2}	{-1, 0, 1}	{-1, 0, 1}	{-1, 0, 1}	{-1, 0, 1}	{-1, 0, 1}	{-1, 0, 1}	{-1, 0, 1}	{-1, 0, 1}
	-0.14	0.95	1.02	1.03	1.06	0.82	1.04	0.79	0.95
$-u_H/u_C$	[-0.54, 0.27]	[0.57, 1.27]	[0.80, 1.21]	[0.82, 1.22]	[0.83, 1.25]	[0.41, 1.08]	[0.79, 1.24]	[0.26, 1.10]	[0.69, 1.17]
	0.18	0.58	0.63	0.64	0.66	0.69	0.62	0.62	0.68
$u_{C_{\text{others}}}/u_H$	[-0.07, 0.43]	[0.43, 0.75]	[0.55, 0.74]	[0.54, 0.73]	[0.56, 0.76]	[0.58, 0.80]	[0.51, 0.72]	[0.49, 0.76]	[0.59, 0.79]
	0.01	0.34	0.29	0.29	0.24	0.25	0.29	0.35	0.21
$u_{C_{\text{others}}}/u_C$	[-0.54, 0.49]	[0.06, 0.67]	[0.08, 0.53]	[0.09, 0.52]	[0.00, 0.48]	[-0.08, 0.56]	[0.02, 0.53]	[-0.15, 0.96]	[-0.06, 0.49]
	-0.05	-0.20	-0.18	-0.19	-0.16	-0.18	-0.18	-0.21	-0.14
$u_H/u_C + u_{C_{\text{others}}}/u_C$	[-0.41, 0.28]	[-0.47, 0.03]	[-0.34, -0.05]	[-0.34, -0.06]	[-0.32, 0.00]	[-0.39, 0.05]	[-0.34, -0.01]	[-0.57, 0.09]	[-0.34, 0.04]
	-0.28	-0.79	-0.82	-0.83	-0.81	-0.84	-0.80	-0.81	-0.82
	[-0.72, 0.19]	[-1.14, -0.51]	[-1.02, -0.66]	[-1.01, -0.65]	[-1.01, -0.62]	[-1.13, -0.61]	[-1.01, -0.59]	[-1.21, -0.49]	[-1.06, -0.62]
Observations	1,264	847	2,097	2,097	1,367	1,366	1,706	707	1,335

Notes: Column 1: difference between estimates under the \$5,000 and \$3,000 baselines. Other columns: parameter estimates. Column 2: respondents within \$1,000 of the baseline spending level. Column 3: finite-horizon model. Column 4: nonzero, wave-varying response-error means. Column 5: first wave only. Column 6: respondents whose internal-habit rationales match the ceteris-paribus scenario. Columns 7-9: additional attention checks. Column 7 excludes respondents failing either instruction manipulation check. Column 8 excludes respondents with inconsistent past-spending reports in the time-discount-rate question. Column 9 excludes respondents with at least one polar response. 99% CIs in brackets. $\text{sgn}(\cdot) \in \{-1, 0, +1\}$; its 99% CI is the set of sign values not rejected at the 1% level.

1 for reasonable values of T , the finite time horizon of interest.³ Consequently, estimates under the finite horizon yield parameter values (Column 3 of Table A.4) that are virtually indistinguishable from the benchmark estimates under the infinite horizon.

D.3 Response Bias and Error with Nonzero and Wave-Varying Mean

The statistical model assumes a zero mean for the response error across both waves of the survey. By relaxing this assumption, we obtain a statistical model that allows response biases and errors to have nonzero means, which may also differ across waves. Without loss of generality,⁴ the joint distribution of parameter $\tilde{x}$ for individual i in both waves becomes

$$\begin{bmatrix} \tilde{x}_{i,1} \\ \tilde{x}_{i,2} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mu_x + \mu_\varepsilon \\ \mu_x - \mu_\varepsilon \end{bmatrix}, \begin{bmatrix} \sigma_x^2 + \sigma_{\varepsilon_x,P}^2 + \sigma_{\varepsilon_x,T}^2 & \sigma_x^2 + \sigma_{\varepsilon_x,P}^2 \\ \sigma_x^2 + \sigma_{\varepsilon_x,P}^2 & \sigma_x^2 + \sigma_{\varepsilon_x,P}^2 + \sigma_{\varepsilon_x,T}^2 \end{bmatrix} \right).$$

The estimates of the means of the response biases and errors in wave one (or, equivalently, the negatives of the means of the response biases and errors in wave two) are indistinguishable from zero at the 99%

³The survey instructs respondents: “If easier, think of ... ‘Future’ as the next 30 years.” Accordingly, T is set to 30 in the finite-horizon estimation. Estimates are similarly insensitive to alternative reasonable values of T (e.g., 10 and 20).

⁴Only two means, one for each wave, can be identified. The specification here, which identifies the average and the difference of the means, is equivalent to a specification that uses two parameters, one for each mean. If the two means are different, μ_ε should be statistically significantly different from 0.

level for all parameters of interest. This implies that the parameter estimates are robust to response biases and errors with nonzero, wave-varying means, as confirmed by Column 4 of Table A.4.

D.4 First-Wave-Only Estimation

The benchmark estimation uses responses from both waves of the survey. To assess whether the two-wave design itself drives the headline results, we re-estimate the model using only first-wave responses. The estimates (Column 5 of Table A.4) are quantitatively close to the benchmark, indicating that the second wave serves to mitigate response error and to enable the test-retest validity check, rather than to determine the main parameter estimates.

D.5 Strict Ceteris-Paribus Compliance

The benchmark sample already screens out respondents who fail the comprehension checks on the hypothetical scenario—in particular, those who do not correctly identify that spending is the only dimension that varies across the two “parallel universes,” with all other factors held fixed. Only respondents who answer this check correctly are admitted into the main sample: 93% pass on the first attempt and the remaining 7% pass on their second attempt, so every respondent in the benchmark sample has explicitly demonstrated comprehension of the ceteris-paribus structure. Even after passing the comprehension checks, a respondent may implicitly violate the ceteris-paribus structure (for example, by treating higher future spending as coinciding with higher inflation, or a lower spending path as leaving room for future saving). To gauge the impact of this, we restrict the sample to respondents whose open-ended rationale for the existence-of-internal-habit question explicitly cites the change in own spending relative to their past experience as the reason for their satisfaction comparison, indicating that their reasoning did not violate the ceteris-paribus assumption. This restriction drops any respondent whose rationale falls outside this pattern, leaving 905 respondents in wave one and 461 in wave two (1,366 observations in total). Estimates from this strictly compliant subsample (Column 6 of Table A.4) preserve every qualitative conclusion of the paper. The point estimate of the annual habit decay rate θ moves to 0.82, but its 99% confidence interval, [0.41, 1.08], includes the benchmark estimate of 1.03, so the data remain consistent with a fast habit-decay process in this subsample as well. The conclusions of Section 4.1 are likewise unaffected:

even at the interval’s lower bound, $\theta = 0.41$ implies an annual decay factor of 0.34, above the 0.30 used in Column 4 of Table 6, at which the model-implied moments already deviate from reality by at least 40%.

D.6 Additional Attention Checks

In fielding the survey, explicit attention checks were used to exclude respondents who failed to comprehend the hypothetical scenario or monthly spending profiles, or who did not correctly respond to two instruction manipulation checks—questions prompting specific responses. This subsection employs additional attention checks to investigate whether undetected lapses in respondent attention, not captured by the default measures, may introduce bias into the estimates.

In the benchmark sample, respondents failing both instruction manipulation checks are excluded. Excluding respondents who fail either check results in a reduction of the sample size by 247 in the first wave and 144 in the second wave. Estimates derived from this subsample (Column 7 of Table A.4) are essentially the same as those from the main sample.

A third supplementary attention check evaluates whether respondents exhibit indifference between response options that are objectively the same. In the time discount rate question (see Appendix E.1), past spending profiles are identical across the two universes, requiring respondents to report consistent past experiences. Excluding respondents who provided inconsistent responses substantially reduces the sample by 1,390 observations across both waves. This significant reduction may be attributed to the unique design of this question, which is the only question throughout the survey that includes a response option reflecting identical past spending profiles. Despite the increased standard errors resulting from the smaller sample size, the estimates (Column 8 of Table A.4) remain closely aligned with the baseline estimates.

Finally, we employ a measure of response consistency across waves as an additional attention check. Given its speculative nature, this check excludes only those respondents who provided at least one polar response—defined as selecting the first (or last) bracket of parameter values in wave one and the last (or first) in wave two. This criterion results in the exclusion of 762 responses across both waves combined. Nevertheless, the estimates derived from this filtered sample (Column 9 of Table A.4) remain virtually identical to the baseline estimates. Collectively, these attention checks support the robustness of this study’s core findings against potential response inattention.



Figure A.1: Spending Profiles for a Survey Question on Time Discount Rate

E Additional Survey Questions

This appendix presents the supplementary survey questions referenced in the main text, including those related to the time discount rate and the annual habit decay factor of 0.9, along with an analysis of scale-use tendencies derived from calibration questions.

E.1 Time Discount Rate

To elicit the time discount rate, the survey presents respondents with a trade-off between a larger spending increase in the subsequent year and a smaller spending increase sustained across all future years. The corresponding survey question features spending profiles as illustrated in Figure A.1. Choosing the larger spending increase in the next year indicates greater impatience, which explains the direction of the inequality in Proposition 4.

Proposition 4. *Under a first-order approximation, $\rho > -\ln \left(1 - \frac{\Delta C_2}{\Delta C_1}\right)$ if a respondent chooses Universe One over Universe Two for a better future experience in a time discount rate question.⁵*

Proof. A respondent preferring Universe One over Universe Two for a better future experience in the time discount rate question implies

$$\begin{aligned}
0 &< U(\text{Universe One}) - U(\text{Universe Two}) \\
&= [U(\text{Universe One}) - U(\text{Baseline})] - [U(\text{Universe Two}) - U(\text{Baseline})] \\
&= \Upsilon(0, \Delta C_1) - \Psi(0, \Delta C_2) \\
&= \int_0^1 e^{-\rho t} [u_C \Delta C_1 + u_H (1 - e^{-\theta t}) \Delta C_1] dt + \int_1^\infty e^{-\rho t} [u_H e^{-\theta t} (e^\theta - 1) \Delta C_1] dt \\
&\quad - \int_0^\infty e^{-\rho t} [u_C \Delta C_2 + u_H (1 - e^{-\theta t}) \Delta C_2] dt \\
&= [\Delta C_1 (1 - e^{-\rho}) - \Delta C_2] \frac{1}{\rho} \left(u_C + u_H \frac{\theta}{\rho + \theta} \right),
\end{aligned}$$

where the third equality holds under a first-order approximation.

Since $u_C > 0$ and $u_C > -u_H$ (equivalently $u_C + u_H > 0$) hold under the estimates of the relevant parameters, we have

$$u_C + u_H \frac{\theta}{\rho + \theta} = \left(1 - \frac{\theta}{\rho + \theta}\right) u_C + \frac{\theta}{\rho + \theta} (u_C + u_H),$$

which is a convex combination of two positive numbers and is therefore positive. This implies

$$\Delta C_1 (1 - e^{-\rho}) - \Delta C_2 > 0,$$

or equivalently

$$\rho > -\ln \left(1 - \frac{\Delta C_2}{\Delta C_1}\right).$$

□

Response frequencies for sequences U2U2, U2U1, U1U2, U1U1 were 57.3%, 9.2%, 12.7%, 20.8% in the first wave and 55.3%, 8.1%, 11.6%, 24.9% in the second wave, respectively. Estimation yields

⁵In Figure A.1, $\Delta C_1 = \$2,000$ and $\Delta C_2 = \$200$. Table A.5 provides additional values of ΔC_1 and ΔC_2 used in the survey.

Table A.5: Quantities and Thresholds for Spending Profiles for Time Discount Rate

Quantity and threshold	Baseline = \$5,000			Baseline = \$3,000		
	Choosing U2	Initial question	Choosing U1	Choosing U2	Initial question	Choosing U1
ΔC_1	2000	2000	2000	1000	1000	1000
ΔC_2	100	200	500	50	100	300
Threshold	0.05	0.11	0.29	0.05	0.11	0.36

Notes: Quantities and implied thresholds mentioned in Proposition 4. “Choosing U1” and “Choosing U2”: follow-up questions shown after the initial Universe One or Universe Two choice. Quantities are in U.S. dollars.

a time discount rate of 0.10, with a standard error of 0.01, which aligns with the conventional range of experimental estimates for this parameter (Matousek et al., 2022).

E.2 Another Habit Decay Speed Question

Figure A.2 presents the spending profiles from the survey question designed to examine the relationship between 0.9 and the (annual) habit decay factor.

E.3 Scale-Use Tendency

To assess scale-use tendencies, the survey incorporates nine visual calibration questions developed by Benjamin et al. (2023), which evaluate the perceived darkness of a circle, shape complexity, and a hypothetical individual’s confidence level. For instance, Figure A.3a illustrates a question prompting respondents to assess the darkness of a circle. Appendix F provides screenshots of all survey questions, including those related to shape complexity and confidence. Although these calibration questions—selected for their rapid response times—may not fully capture scale use in the happiness dimension, Benjamin et al. (2023) demonstrate their effectiveness in measuring general scale-use tendencies. Following the framework of Benjamin et al. (2023), we derive respondent-level scale-use parameters—namely, the shifter and stretcher—based on responses to these questions, which were administered both before and after the core survey modules. Figures A.3b and A.3c present scatterplots of the scale-use parameters across both administrations, demonstrating a high level of consistency in scale-use tendencies between the beginning and end of the survey.

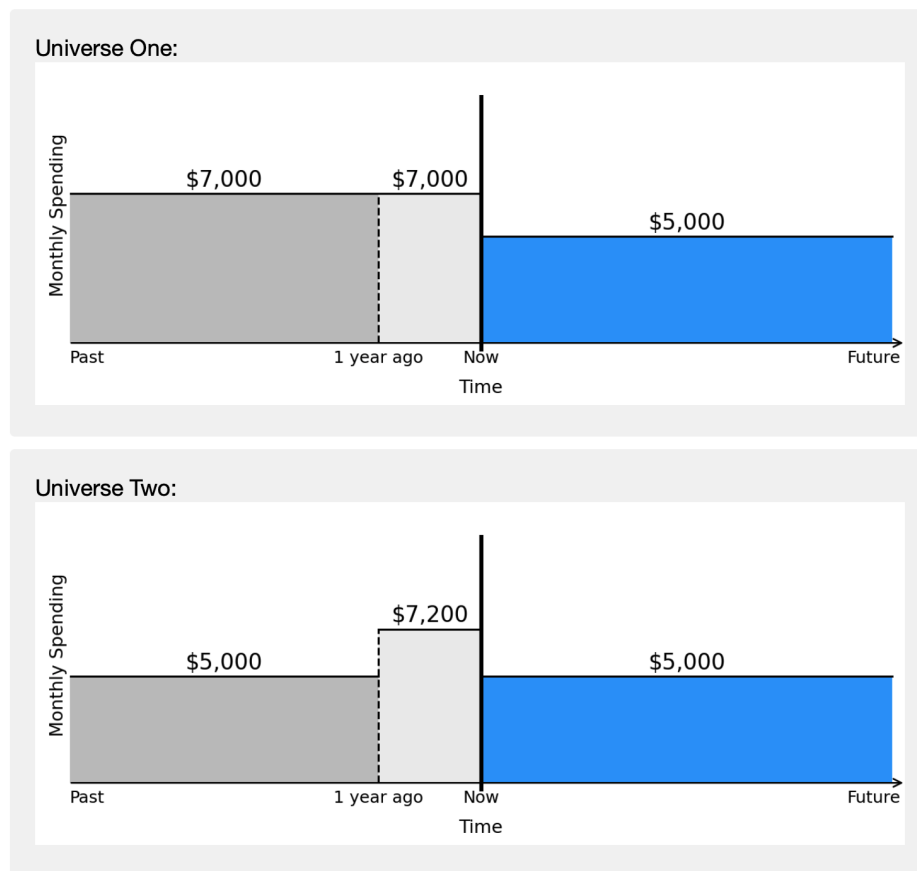
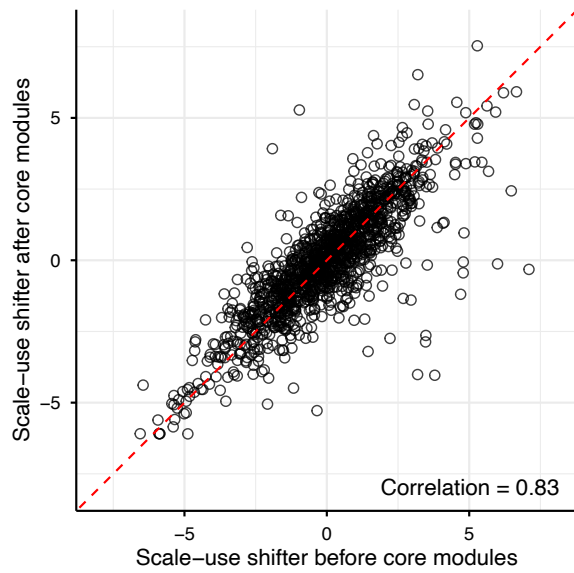


Figure A.2: Spending Profiles for Relationship Between (Annual) Habit Decay Factor and 0.9

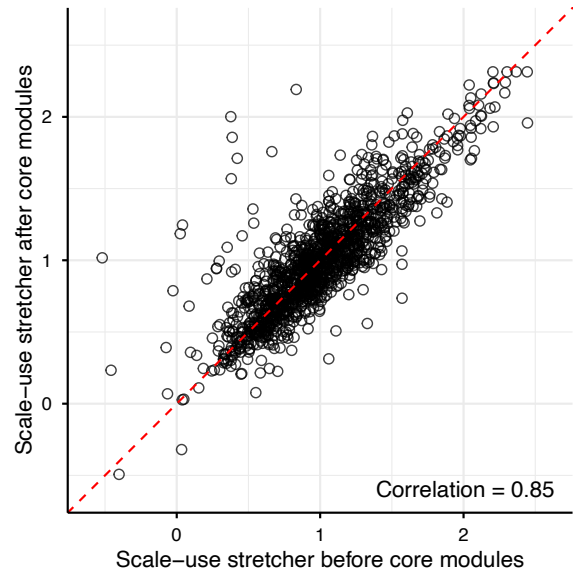
Please rate **how dark each of the following circles looks** on a scale from 0 to 10, where **0 means “Lowest level of darkness possible”** and **10 means are “Highest level of darkness possible”**.



(a) An example calibration question



(b) Scale-use shifter before and after core modules



(c) Scale-use stretcher before and after core modules

Figure A.3: Example Calibration Question and Consistency of Scale-Use Tendencies

Notes: Panel (a): calibration question asking respondents to rate the darkness of a circle on a slider. Panels (b) and (c): scatterplots of scale-use parameters before and after the core modules. Red lines: 45-degree reference lines.

Appendix-Specific References

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